Hamiltonian is just $H = p_x^2/2m$, where m is the particle's mass, so according to Axiom V and Eq. (5.145),

$$\Psi(x, t) = e^{-ip_x^2 t/2m\hbar} \frac{1}{\sqrt[4]{\pi a^2}} e^{-x^2/2a^2} .$$

Using Eq. (5.146) and the spectral resolution which we have already determined for p_x we have immediately

$$\Psi(x, t) = \frac{1}{\sqrt[4]{\pi a^2}} \int_{-\infty}^{\infty} e^{i\hbar k^2 t/2m} \, d[E(k)\, e^{-x^2/2a^2}] .$$

But according to our previous work,

$$E(k)\, f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{k} e^{ik'x} \hat{f}(k')\, dk' ,$$

where $\hat{f}$ denotes the Fourier transform of f. The Fourier transform of $\Psi(x, 0)$ is easily computed; we find

$$\hat{\Psi}(k, 0) = \sqrt[4]{\frac{a^2}{\pi}}\, e^{-a^2k^2/2} .$$

Thus

$$\Psi(x, t) = \sqrt[4]{\frac{a^2}{4\pi^3}} \int_{-\infty}^{\infty} e^{-i\hbar k^2 t/2m} \, d\left[\int_{-\infty}^{k} e^{ik'x} e^{-a^2k'^2/2}\, dk'\right]$$

$$= \sqrt[4]{\frac{a^2}{4\pi^3}} \int_{-\infty}^{\infty} e^{-i\hbar k^2 t/2m} e^{-a^2k^2/2} e^{ikx}\, dk .$$

This last integral is readily evaluated. We find that

$$\Psi(x, t) = \frac{1}{\pi^{1/4}} \frac{1}{\sqrt{a + i\dfrac{ht}{ma}}} \exp\left[-\frac{x^2}{2a(a + iht/ma)}\right] ,$$

and therefore the probability density is given by

$$\rho(x, t) = |\Psi(x, t)|^2 = (1/\sqrt{\pi\Delta^2})\, e^{-x^2/\Delta^2} ,$$

where $\Delta^2 = a^2 + (ht/ma)^2$. Thus the width of the probability distribution increases in time. This phenomenon is known as "the spreading of the wave packet." The initial width was equal to $2a$; the width at time t is $2\sqrt{a^2 + (ht/ma)^2}$. To get a feeling for the magnitudes involved, suppose that we are dealing with electrons and that $a \equiv 1$ mm. This might describe a case in which one creates a beam of electrons collimated by a slit of width approximately equal to one millimeter. Then if we use $m = 9 \times 10^{-28}$ gm and $h = 10^{-27}$ erg-sec, we find that the width has doubled in about 15 msec. This is a very short time on a macroscopic scale, but on a scale appropriate to electrons, it is a very large time. For example, a velocity of 10^8 cm/sec is not unusual for an electron; in 15 msec, such an electron

would travel 1.5×10^6 cm, or 15 km! Thus a beam of electrons initially having a width of 1 mm and moving with a velocity of 10^8 cm/sec would travel 15 km before doubling in width. This assumes, of course, that the electrons are truly free; i.e., we neglect collisions, electromagnetic forces, gravitational forces, etc.

Before we can claim to have a complete system of axioms, we must give some method by which one can gain information about the state vector Ψ that figures so prominently in the previous axioms. The key to this problem, which is clearly of paramount practical importance, is the following axiom.

Axiom VI. If at time $t = 0$ one measures simultaneously the commuting observables A, B, and C, and finds with certainty that the values of these observables lie between a_1 and a_2, b_1 and b_2, and c_1 and c_2 respectively, then immediately after the measurement, the state vector of the system satisfies the relation

$$[E_A(a_2) - E_A(a_1)][E_B(b_2) - E_B(b_1)][E_C(c_2) - E_C(c_1)]\,\Psi = \Psi\,. \quad (5.148)$$

We may say that the measurements project the original wave function into the subspace of Hilbert space associated with the projection operator

$$[E_A(a_2) - E_A(a_1)][E_B(b_2) - E_B(b_1)][E_C(c_2) - E_C(c_1)]\,.$$

Note that because of Axiom IV, Eq. (5.148) implies that if we remeasure A (or B or C) immediately after our first measurement, we will find a value between a_1 and a_2 (or b_1 and b_2, or c_1 and c_2) with certainty.

Let us first consider the measurement of a single observable, say A. If there is only one point, a, in the spectrum of A between a_1 and a_2, then according to Eq. (5.141),

$$\Psi = e^{i\alpha}\Phi_a\,,$$

if a is nondegenerate, when Φ_a is the normalized eigenvector of A belonging to a and $e^{i\alpha}$ is a phase factor which is indeterminate. Clearly, according to our previous axioms, this phase factor can have no observable consequences. In this case, the measurement of A has, so to speak, "forced" the system into an eigenstate of A. Note that this does *not* tell us that the system was in the state Φ_a *before* the measurement, but only that it is in the state Φ_a *after* the measurement. We have "prepared" the system in the state Φ_a. If the eigenvalue a has multiplicity $\mu_a > 1$, then according to Eq. (5.142), after the measurement of A, we can only say that the state vector lies in the subspace of Hilbert space spanned by the μ_a orthonormal eigenvector of A belonging to a, that is,

$$\Psi = \sum_{\nu=1}^{\mu_a} c_\nu \Phi_a^{(\nu)}\,,$$

where

$$\sum_{\nu=1}^{\mu_a} |c_\nu|^2 = 1\,.$$

Thus in this case we have not completely specified the state vector by measuring A.

However, according to Axiom VI, we have additional resources at our disposal; we can choose *another* observable, B, whose self-adjoint operator commutes with A, and measure A and B simultaneously. Suppose we discover that the value of A lies with certainty between a_1 and a_2 and the value of B lies with certainty between b_1 and b_2. Again, we imagine that a is the only point in the spectrum of A between a_1 and a_2 and that a has multiplicity μ_a. Hence

$$[E_A(a_2) - E_A(a_1)]\Psi = \sum_{\nu=1}^{\mu_a} c_\nu \Phi_a^{(\nu)} . \tag{5.149}$$

We assume also that there is only one point, b, between b_1 and b_2 in the spectrum of B, and that b has multiplicity μ_b. Using Eqs. (5.142) and (5.149), we find that

$$[E_B(b_2) - E_B(b_1)][E_A(a_2) - E_A(a_1)]\Psi = \sum_{\nu=1}^{\mu_a} \sum_{\nu=1}^{\mu_b} c_\nu (\Phi_b^{(\lambda)}, \Phi_a^{(\nu)})\, \Phi_b^{(\lambda)} .$$

We know from Theorem 4.22 that every eigenvector of A must be an eigenvector of B *and vice versa*, so $(\Phi_b^{(\lambda)}, \Phi_a^{(\nu)}) = 1$ if $\Phi_b^{(\lambda)} = \Phi_a^{(\nu)}$, and vanishes otherwise. It may happen that there is only one pair, $(\bar{\lambda}, \bar{\nu})$, for which $(\Phi_b^{(\lambda)}, \Phi_a^{(\nu)}) = 1$. In this case, Axiom VI tells us that

$$\Psi = e^{i\beta}\Phi_b^{(\bar{\lambda})} = e^{i\beta}\Phi_a^{(\bar{\nu})} ,$$

so that we have uniquely specified the state vector of the system.

If there is more than one pair for which $(\Phi_b^{(\lambda)}, \Phi_a^{(\nu)})$ fails to vanish, then calling these pairs (λ_i, ν_i), we have

$$\Psi = \sum_i c_{\nu_i} \Phi_b^{(\lambda_i)} = \sum_i c_{\nu_i} \Phi_a^{(\nu_i)} ;$$

that is, Ψ lies in the subspace of Hilbert space spanned by the mutual eigenvectors of A and B belonging to a and b. In this last case, the state vector is not uniquely determined, so we must find a new observable, C, which commutes with A and B, and repeat the process. In this manner we eventually will specify the state uniquely.

A collection of commuting observables which has the property that simultaneous measurement of the observables specifies uniquely the state vector of a system is called a *complete set* of commuting observables. For example, if we consider an electron in a coulomb field and neglect electron spin, then H, L^2, and L_z (where $H = p^2/2m - e^2/r$ and $\mathbf{L}$ is the orbital angular momentum of the electron) constitute such a set; H and L^2 alone do not constitute a complete set of observables, for after a measurement of the energy and the total angular momentum, there is still a degeneracy corresponding to the possible values of the z-component of the angular momentum. If we include the spin of the electron, then H, L^2, S^2, J^2, and J_z constitute a complete set, where $\mathbf{S}$ is the electron's spin angular momentum, and $\mathbf{J} = \mathbf{L} + \mathbf{S}$ is the total angular momentum

of the electron. This set is not unique; H, L^2, L_z, S^2 and S_z would do just as well. Which set is more useful depends on the practical situation. If we wish to take into account relativistic effects by including magnetic interactions via a term in the Hamiltonian of the form $A\mathbf{L}\cdot\mathbf{S}/r^3$, then the second set mentioned above is no longer a complete commuting set, whereas the first is (because

$$\mathbf{L}\cdot\mathbf{S} = (J^2 - L^2 - S^2)/2) .$$

Thus Axiom VI enables us to perform the all-important task of controlling the initial conditions of an experiment by the process of preparing quantum states. Without this axiom, Axiom IV would have no content, since if one never knew what the state vector of a system was, one could never hope to compute the probability of obtaining a particular result in a measurement.

PROBLEMS

1. Prove that $\int_{-1}^{1} P_n(x)dx = 0$ for $n \neq 0$ in two ways.
2. A metal spherical shell of radius R with its center at $x = y = z = 0$ is cut in half along its intersection with the plane $z = 0$. The two halves are separated by an infinitesimal gap and the upper and lower hemispheres are brought to voltages $+V$ and $-V$ respectively. Show that the potential inside the sphere is

$$\phi(r, \theta) = V \sum_{l=0}^{\infty} (-1)^l \left(\frac{r}{R}\right)^{2l+1} \frac{(2l)!}{(2^l l!)^2} \frac{4l+3}{2l+2} P_{2l+1}(\cos\theta) ,$$

where θ is the polar angle measured relative to the positive z-axis.
Hint: $\nabla^2\phi = 0$ inside and on a spherical shell. ϕ is clearly independent of ϕ. Following the usual procedure of separation of variables, we find that the solution is of the form

$$\phi(r, \theta) = \sum_{l=0}^{\infty} [A_l r^l + B_l r^{-l-1}][C_l P_l(\cos\theta) + D_l Q_l(\cos\theta)] .$$

Immediately set $B_l = 0$ (all l) so $\phi(0, \theta)$ is finite and set $D_l = 0$ (all l) so $\phi(r, 0)$ and $\phi(r, \pi)$ are finite. The problem is now one of mathematics—determining the constants $A_l C_l \equiv \alpha_l$ such that the boundary conditions are satisfied:

$$\phi(R, \theta) = \begin{cases} +V & \text{for} \quad 0 \leq \theta < \pi/2 , \\ -V & \text{for} \quad \pi/2 < \theta \leq \pi . \end{cases}$$

3. A conducting sphere of radius d sits in a charge-free region in an electric field. Its surface is kept at a fixed distribution of electric potential $V = F(\theta)$, where (r, ϕ, θ) are spherical coordinates with the origin at the center of the sphere. Determine the potential at all points in the space inside and outside the sphere; i.e., solve Laplace's equation $\nabla^2 V = 0$, assuming the following boundary conditions:

   i. $\lim_{r\to d} V(r, \theta) = F(\theta), \qquad 0 < \theta < \pi ,$

   ii. $\lim_{r\to\infty} V(r, \theta) = 0$

   (potential vanishes at points infinitely far away from sphere).

Hint: Use the method of separation of variables. Solve the r equation by making the substitution $r = e^t$. When it is solved, set the exponent $-\frac{1}{2} + (\lambda + \frac{1}{4})^{1/2} = n$ so that $\lambda = n(n+1)$ (λ = separation constant). Solve the θ equation by an appropriate change of variable.

Answer:

$$\textit{Inside sphere}: \quad V(r,\theta) = \sum_{n=0}^{\infty} \frac{2n+1}{2} \frac{r^n}{d^n} P_n(\cos\theta) \int_{-1}^{1} F(x) P_n(x)\, dx$$

$$r \leq d, \; F(\theta) = f(\cos\theta)\,.$$

$$\textit{Outside sphere}: \quad V(r,\theta) = \sum_{n=0}^{\infty} \frac{2n+1}{2} \frac{d^{n+1}}{r^{n+1}} P_n(\cos\theta) \int_{-1}^{1} F(x) P_n(x)\, dx\,.$$

4. a) Show that the 2π periodic Fourier series representation of the square wave (step function)

$$f(x) = \begin{cases} -1, & -\pi < x < 0\,, \\ +1, & 0 < x < \pi\,, \end{cases} \quad \text{is} \quad f(x) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin(2n-1)x\,.$$

Compare this result with the expansion of the same function in Legendre polynomials—see Jackson, *Classical Electrodynamics*, pp. 58–59.

b) Derive from the Fourier series obtained in (a) a simple infinite series expression for π; preferably one with rapid convergence. It is instructive to plot several of the approximations to $f(x)$; that is, plot

$$S_m(x) = \frac{4}{\pi} \sum_{n=1}^{m} \frac{1}{2n-1} \sin(2n-1)x$$

for several values of m to see how $S_m(x) \to f(x)$ with increasing m.

c) *Gibbs' phenomenon*: Show that the Fourier series overshoots the function just to the right and left of the origin by about 18%. This is not easy. Guidance may be found in Sommerfeld's *Partial Differential Equations in Physics*, pp. 7–12, or in Morse and Feshbach, p. 747.

5. Prove that if $f(x)$ is real,

$$f(x) = \frac{1}{\pi} \int_0^{\infty} du \int_{-\infty}^{\infty} f(t) \cos u(t-x)\, dt\,.$$

6. Solve the integral equation

$$\int_0^{\infty} g(t) \cos xt\, dt = \begin{Bmatrix} e^{-\alpha x} & \text{for } x > 0 \\ e^{\alpha x} & \text{for } x < 0 \end{Bmatrix} \qquad (\alpha > 0)\,.$$

Answer:

$$g(t) = \frac{2}{\pi} \frac{\alpha}{\alpha^2 + t^2}\,.$$

7. a) Show that the energy transferred to a spring by a time-dependent force $F(t)$ defined by (Fig. 5.6)

$$F(t) = \begin{Bmatrix} 0, & t < -\tau \\ F_0(t/2\tau + 1/2), & |t| \leq \tau \\ F_0, & t > \tau \end{Bmatrix} \quad \text{is} \quad \Delta E = \frac{1}{2m} (F_0/\omega)^2 \left(\frac{\sin \omega\tau}{\omega\tau}\right)^2.$$

Hint: Take derivatives of $F(t)$ until you reach a function whose Fourier transform is known.

b) Show that the energy transfers in the adiabatic and sudden limits agree with these limits as determined for the probability integral force function (Example 3, Section 5.7).

c) For certain finite values of $\omega\tau$ there will be no energy transfer to the spring. Determine these values and explain their origin physically.

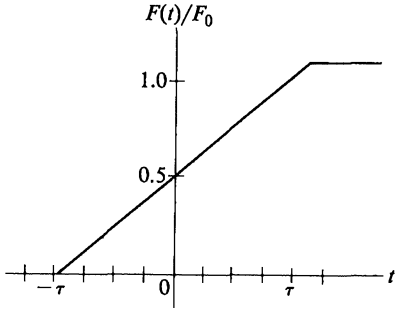

Fig. 5.6.

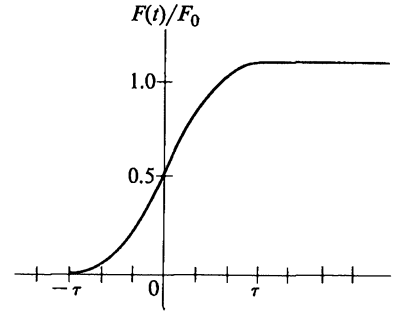

Fig. 5.7.

8. Work Problem 7 for the force function (Fig. 5.7)

$$F(t) = \begin{cases} 0\ , & t < -\tau\ , \\ F_0\left[\frac{1}{2} + \frac{1}{4}\left(\left(\frac{3t}{\tau}\right) - \left(\frac{t}{\tau}\right)^3\right)\right], & |t| \leqslant \tau\ . \\ F_0\ , & t > \tau\ . \end{cases}$$

Answer:

$$\Delta E = \frac{1}{2m}(F_0/\omega)^2 \frac{9}{(\omega_0\tau)^6}(\sin\omega_0\tau - \omega_0\tau\cos\omega_0\tau)^2\ .$$

The adiabatic and sudden limits must and do (prove this) agree with those of Problem 7. For part (c), settle for a determination of the position of the first node of ΔE. You should get

$$\omega\tau \cong 4.49\ .$$

9. Compute the three-dimensional Fourier transforms of the following functions:

a) $H(\mathbf{r}) = \dfrac{e^{-\lambda|\mathbf{r}|}}{|\mathbf{r}|}$

b) $F(\mathbf{r}) = \dfrac{1}{|\mathbf{r}|}$ [To save work consider $F(\mathbf{r})$ as a special case of $H(\mathbf{r})$.]

c) $Y(\mathbf{r}) = e^{-\lambda|\mathbf{r}|}$ [The Fourier transform of this function can be obtained from that of $H(\mathbf{r})$ also.]

10. Maxwell's equations in free space in Gaussian units are:

$$\text{curl}\ \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{H}}{\partial t}, \qquad \text{div}\ \mathbf{H} = 0\ , \qquad \text{curl}\ \mathbf{H} = \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t}, \qquad \text{div}\ \mathbf{E} = 0\ .$$

At any time t the electric (and magnetic) fields may be Fourier analyzed:

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{(2\pi)^{3/2}} \int_{-\infty}^{\infty} \mathbf{E}(\boldsymbol{k}, t) e^{i\mathbf{k}\cdot\mathbf{r}}\, d\mathbf{k}, \quad \text{etc.}$$

[Here the Fourier components of $\mathbf{E}(\mathbf{r}, t)$ have been written $\mathbf{E}(\mathbf{k}, t)$; the argument identifies $\mathbf{E}(\mathbf{k}, t)$ as the Fourier component of the field, distinguishing it from the field itself, which has argument $\mathbf{r}$. If you don't like this notation, adopt another.]

a) Prove that the Fourier components of the electric and magnetic fields satisfy the following equations:

   i) $\dot{\mathbf{H}}(\mathbf{k}, t) = -ic(\mathbf{k} \times \mathbf{E}(\mathbf{k}, t))$
   ii) $\mathbf{k}\cdot\mathbf{E}(\mathbf{k}, t) = 0$
   iii) $\dot{\mathbf{E}}(\mathbf{k}, t) = ic(\mathbf{k} \times \mathbf{H}(\mathbf{k}, t))$
   iv) $\mathbf{k}\cdot\mathbf{H}(\mathbf{k}, t) = 0$.

   These four equations are equivalent to Maxwell's equations. This reformulation is well-suited as a starting point for quantum mechanical considerations of the electromagnetic field.

b) Using the results of part(a), prove that the Fourier components of the electric field satisfy the homogeneous wave equation. That is, prove

$$\left[\frac{\partial^2}{\partial t^2} + c^2k^2\right]\mathbf{E}(\mathbf{k}, t) = 0\,.$$

11. Show informally that

$$\lim_{g\to\infty} \frac{\sin gx}{\pi x}, \qquad g > 0\,,$$

is a representation of the Dirac δ-function.

12. Prove that

$$\delta(x) = \begin{cases} \lim\limits_{n\to\infty}\left(\dfrac{1}{|a|} c_n \left[1 - \left(\dfrac{x}{a}\right)^2\right]^n\right) & \text{if} \quad 0 \le |x| \le |a|\,, \\ 0 & \text{otherwise} \end{cases}$$

is also a representation of the δ-function, where c_n is given in Eq. (5.28). Show, therefore, that $\delta(ax) = (1/|a|)\,\delta(x)$.

13. Show that if $r = (x^2 + y^2 + z^2)^{1/2}$, then

$$\nabla^2\left(\frac{1}{r}\right) = -\,4\pi\delta^3(\mathbf{r})$$

[*Hint*: Use Green's Theorem.]

14. Prove Parseval's equation (5.11) in Hilbert space by applying Bessel's completeness relation (5.10) to the function $f + g$ and then subtracting the corresponding equations for f and g.

15. The parity operator P plays a fundamental role in many areas of physics. Show that the Gegenbauer polynomials are eigenfunctions of P; that is, if the leading power of x in $G_n^m(x)$ is even, then $G_n^m(x)$ contains *only* even powers of x. One way to do this is to assume the contrary. Write $G_n^m(x) = g_n^m(x) + u_n^m(x)$, where $g_n^m(x)$ contains only even powers of x and $u_n^m(x)$ contains only odd powers, and conclude that $(L - \lambda_n)g_n^m = 0$ and $(L - \lambda_n)u_n^m = 0$, where L is the Sturm-Liouville operator which has the Gegenbauer polynomials as eigenfunctions. Thus $g_n^m(x)$ and $u_n^m(x)$

are linearly independent (because they belong to different eigenvalues of P) solutions of the Gegenbauer equation, and they are both polynomial solutions. From this conclude that $u_n^m(x) = 0$. [*Hint*: we know that for the Gegenbauer polynomials $\lambda_n = -n(n + 2m + 1)$.] Thus $G_n^m(x)$ contains only even powers of x and has parity eigenvalue $+1$.

16. Prove that the complete set of orthonormal functions on the unit sphere, Y_{lm}, is unique up to a phase factor. Use induction on l for $m = m'$ fixed.
17. If $[x_i, p_j] = i\hbar\delta_{ij}$, where the indices run from one to three, and if we define $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, then show that

$$[L_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k .$$

Defining $L_+ = L_x + iL_y$ and $L_- = L_x - iL_y$, show that $[L_+, L_-] = 2\hbar L_z$. Show that if we define

$$\phi_m = L_+^{l+m}\phi_{-l}, \qquad L_-\phi_{-l} = 0 ,$$

where l is an integer, then

$$L_-\phi_m = \hbar(l + m)(l - m + 1)\phi_{m-1} .$$

18. $H_n(x)$ are the Hermite polynomials. Prove for arbitrary n:

a) $H_{2n}(0) = (-1)^n \dfrac{(2n)!}{n!}$, b) $H'_{2n}(0) = 0$,

c) $H_{2n+1}(0) = 0$, d) $H'_{2n+1}(0) = (-1)^n 2\dfrac{(2n+1)!}{n!}$.

19. It should be clear from the definitions that pointwise convergence does not imply uniform convergence. The question comes up, Does pointwise convergence imply convergence in the mean? Show that the sequence of functions

$$f_n(x) = \frac{2n^{1/2}}{(\pi/2)^{1/4}} nxe^{-(nx)^2}$$

converges pointwise for all x to the function zero, that is,

$$\lim_{n\to\infty} f_n(x) = 0 ,$$

but does not converge in the mean to the function zero; i. e.,

$$\lim_{n\to\infty}\int_{-\infty}^{\infty} |0 - f_n|^2\, dx = 1 \neq 0 .$$

20. Let $\{Q_n\}$ be an orthonormal system of Sturm-Liouville polynomials. Prove that the functions $Q'_1, Q'_2, \cdots$ form an orthogonal system of polynomials with weight function $w\alpha$, where the notation is that of Section 5.10.
21. Show that the set of functions $[x^m, x^{m+1}, \cdots]$ is a complete set in the mean on $[-1, +1]$ for $m \geqslant 1$. (Of course, by Weierstrass's theorem, the above is true for $m = 0$.)
22. Obtain the first four orthonormal polynomials for the complete set $[x, x^2, x^3, \cdots]$ on the interval $[-1, +1]$, and write the first few terms of the expansion of $f(x) = 1$ in terms of this set. What do the first two approximations look like graphically?
23. Consider Bessel's equation $Bf_n = n^2 f_n$ on the interval $[0, \infty]$, where

$$B = x^2\frac{d^2}{dx^2} + x\frac{d}{dx} + x^2 .$$

a) Using the methods discussed in Section 5.10, find the weight function appropriate to Bessel's equation.

b) Show that if $n \neq m$ and $w(x)$ is the weight function of Part (a), then

$$\int_0^\infty J_{2n}(x)J_{2m}(x)w(x)\,dx = \int_0^\infty J_{2n+1}(x)J_{2m+1}(x)w(x)\,dx = 0\,.$$

You may make use of the fact that for small x,

$$J_n(x) = \left(\frac{x}{2}\right)^n\left[1 - \frac{x^2}{2(2n+2)} + \cdots\right],$$

while for large x,

$$J_n(x) = \sqrt{\frac{2}{\pi x}}\left[\cos\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right)(1 + O(x^2)) - \frac{4n^2-1}{8}\frac{1}{x}\sin\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right)(1 + O(x^2))\right].$$

Both of these results can be readily derived from complex variable techniques, which will be discussed in Chapter 6.

c) Using the asymptotic formulas given in Part (b), evaluate

$$\int_0^\infty J_{2n+1}(x)J_{2m}(x)w(x)\,dx\,.$$

d) Is the operator, B, in Bessel's equation an Hermitian operator with respect to the space of functions which satisfy

$$\int_0^\infty |f(x)|^2 w(x)\,dx < \infty\,,$$

where $w(x)$ is given by Part (a)? [It is understood, of course, that B can act only on functions f having the property that Bf lies in the Hilbert space if f does.]

24. We have shown that the Legendre polynomials, $P_l(x)$, are solutions to

$$(1 - x^2)f''(x) - 2xf'(x) + l(l+1)f(x) = 0\,, \qquad l = 0, 1, 2, \cdots,$$

which are orthogonal on $[-1, 1]$. However, these are not the only solutions; they are merely the only polynomial solutions.

a) To reassure yourself on this point, show that the function

$$\tanh^{-1} x = \frac{1}{2}\log\left(\frac{1+x}{1-x}\right)$$

satisfies Legendre's equation with $l = 0$. Note that this function is singular at $x = \pm 1$.

b) Show that if we look for a solution of Legendre's equation of the form

$$Q_l(x) = P_l(x)\tanh^{-1} x + \Pi_l(x)\,,$$

then a solution of this form always exists where $\Pi_l(x)$ is a polynomial of degree $l - 1$. Find the inhomogeneous differential equation that $\Pi_l(x)$ must satisfy, and show that this equation always has a solution. What type of boundary condition has been applied in taking $\Pi_l(x)$ to be a polynomial of order $l - 1$?

c) Since $\Pi_l(x)$ is a polynomial of degree $l - 1$ on $[-1, 1]$, it can be written as a linear combination of Legendre polynomials. Find an expression for the co-

efficients in this linear combination. To be precise, show that

$$\Pi_l(x) = 2 \sum_{n=0,1}^{l-1}{}' \frac{2n+1}{(n-l)(n+l+1)} P_n(x) ,$$

where the sum begins at $n = 0$ if l is odd and at $n = 1$ if l is even. The prime on the summation indicates that the sum runs in steps of two.

d) Show that $(Q_l, P_l) = 0$.
[*Hint*: Don't do any integrals.]

e) From their form it is clear that the $Q_l(x)$ lie in the Hilbert space of square-integrable functions on $[-1, 1]$. Do they form an orthogonal set of functions on $[-1, 1]$? Explain.

25. Let us define $E(\lambda)$, an operator-valued function, as in Section 5.11. Denote by λ_n and ϕ_n $(n = 1, 2, \cdots, N)$ the nondegenerate eigenvalues and corresponding eigenvectors of an Hermitian operator, A, and call P_n the projection operator associated with ϕ_n. Then define

$$E(\lambda) = \begin{cases} 0 \text{ for } \lambda < \lambda_1 , \\ \displaystyle\sum_{n=1}^{\nu} P_n \text{ for } \lambda_\nu \leq \lambda < \lambda_{\nu+1} , \qquad \nu = 1, 2, \cdots, N-1 , \\ \displaystyle\sum_{n=1}^{N} P_n = I \text{ for } \lambda \geq \lambda_N . \end{cases}$$

Show that

i) $E(\lambda_1)E(\lambda_2) = E(\lambda_{\min})$, where $\lambda_{\min} = \min(\lambda_1, \lambda_2)$, ii) $E(\lambda)^2 = E(\lambda)$.

26. Let X be the self-adjoint operator on the Hilbert space of square-integrable functions defined by $Xf(x) \equiv xf(x)$.

i) Is X well defined on all elements of the space?
If not, characterize the elements for which it is defined.

ii) Prove that $E_X(\xi)$, defined by

$$E_X(\xi)f(x) = \begin{Bmatrix} f(x) & \text{if } x \leq \xi \\ 0 & \text{if } x > \xi \end{Bmatrix} ,$$

is the resolution of the identity belonging to X.

27. Show that if $E_A(\lambda)$ suffers a discontinuous jump at an eigenvalue $\bar{\lambda}$ of A, then

$$[E_A(\lambda_+) - E_A(\lambda_-)]\psi = \sum_{\nu=0}^{\mu} \Phi_{\bar{\lambda}}^{(\nu)} (\Phi_{\bar{\lambda}}^{(\nu)}, \psi) ,$$

where μ is the multiplicity of $\bar{\lambda}$ and $\{\Phi_{\bar{\lambda}}^{(\nu)}: \nu = 1, 2, \cdots, \mu\}$ is a set of μ orthonormal eigenvectors of A belonging to $\bar{\lambda}$, and λ_+ and λ_- are points on either side of $\bar{\lambda}$ in the intervals of constancy of $E(\lambda)$; that is, $E(\lambda)$ is constant in the intervals $[\bar{\lambda}, \lambda_+]$ and $[\lambda_-, \bar{\lambda}]$. A is any self-adjoint operator.

28. Given that $E_A(\lambda)$ is the resolution of the identity belonging to A, show that

$$[E_A(\lambda_2) - E_A(\lambda_1)]^2 = [E_A(\lambda_2) - E_A(\lambda_1)]$$

if $\lambda_2 > \lambda_1$.

29. i) What is the expectation value of the operator X of Problem 5.26 in the state $\Psi(x) = (1/\sqrt[4]{\pi a^2})e^{-x^2/2a^2}$?
    ii) What is the dispersion, ΔX, about this average value?
    iii) What is the expectation value of $p_x = -ih\, d/dx$ in the above state?
    iv) What is the dispersion, Δp_x, about this average value?

    Note that as the dispersion in X gets smaller (i.e., as a gets smaller, which means that the wave function Ψ becomes more sharply peaked about $x = 0$), the dispersion in p_x gets larger. The product of the two dispersions, $\Delta X \Delta p_x$, is a constant, equal to $\hbar/2$. One can show that for any state Ψ we must always have $\Delta X \Delta p_x \geqslant \hbar/2$.

30. Verify that for all z not in the spectrum of A,

$$(A - zI)^{-1} = \int_{-\infty}^{\infty} \frac{1}{\lambda - z}\, dE_A(\lambda) ,$$

    where A is a self-adjoint operator whose associated spectral resolution is $E_A(\lambda)$.

31. Suppose that we wish to find an eigenvector and an eigenvalue of a self-adjoint operator $A = A_0 + \epsilon A_1$. Imagine that we know the resolution of the identity, $E_{A_0}(\lambda)$, belonging to A_0, and assume that for ϵ sufficiently small any eigenvalue of A can be written as

$$\lambda_n = \lambda_n^{(0)} + \epsilon\lambda_n^{(1)} + \epsilon^2\lambda_n^{(2)} + \cdots$$

    and that the corresponding eigenvector can be written as

$$\phi_n = \phi_n^{(0)} + \epsilon\phi_n^{(1)} + \epsilon^2\phi_n^{(2)} + \cdots$$

    where $A_0\phi_n^{(0)} = \lambda_n^{(0)}\phi_n^{(0)}$. Show in the manner of Section 4.11 that we can find an inhomogeneous equation for $\phi_n^{(1)}$, and that the solution to this equation which satisfies $(\phi_n^{(0)}, \phi_n^{(1)}) = 0$ is

$$\phi_n^{(1)} = -\int_{S_0} \frac{1}{\lambda - \lambda_n^{(0)}}\, d[E_{A_0}(\lambda)(A_1 - \lambda_n^{(1)})\phi_n^{(0)}] , \qquad (1)$$

    where $\lambda_n^{(1)} = (\phi_n^{(0)}, A_1\phi_n^{(0)})$ and S_0 denotes the entire spectrum of A_0 except for an arbitrarily small closed interval about the point $\lambda_n^{(0)}$. We assume that the spectra of A and A_0 are nondegenerate. Show that in the case of a finite-dimensional vector space, the results of Section 4.11 are obtained from Eq. (1).

32. i) Let A be a self-adjoint operator on a Hilbert space H. Assume that the spectrum of A consists of a finite set of points, $\{\lambda_i < 0\colon\ i = 1, 2, \cdots, N\}$, plus the positive real line. Show that if the lowest (i.e., most negative) eigenvalue of A, λ_1, is nondegenerate, then $\lambda_1 = \min\limits_{\psi \in H} (\psi, A\psi)$,
    where ψ is constrained by $||\psi|| = 1$.

    [*Hint*: Note that any vector in H can be written as

$$\psi = c_0\phi_0 + \sum_{i=1}^{N} c_i\phi_i ,$$

    where ϕ_i are the eigenvectors of A, and ϕ_0 is a vector orthogonal to all the ϕ_i $(i = 1, 2, \cdots)$. Thus show that

$$(\psi, A\psi) = \sum_{i=0}^{N} |c_i|^2\lambda_i ,$$

where $\{\lambda_i: i = 1, 2, \cdots\}$ is the set of eigenvalues of A and λ_0 is a *positive* number. Now proceed as in Section 4.9.]

ii) Write down and prove a result analogous to Eq. (1) for λ_2. How would the result generalize for λ_n?

33. Consider the quantum-mechanical observables L_x, L_y, and L_z, which are represented as matrices as follows:

$$L_x = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad L_y = \begin{bmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{bmatrix}, \quad L_z = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

The Hamiltonian for this system has the form $H = H_0 + \alpha L_z$, where H_0 commutes with L_x, L_y, and L_z.

a) Suppose that at $t = 0$ we have prepared the system in an eigenstate of L_z, namely, the state belonging to the eigenvalue $m = +1$ of L_z. If we measure L_z at the later time $t = T$, what is the probability that we will find the value $+1$? the value 0? the value -1?

b) Suppose that instead of measuring L_z at time $t = T$, we measure L_y. What are the possible values of L_x which we can find? What is the probability of finding each of these values?

c) Suppose that at $t = T$ we measure L_y and find with certainty the value -1. After a time τ has elapsed (i.e., at $t = T + \tau$), we remeasure L_y. What is the probability of finding the various allowed values of L_x?

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VOLUME TWO

CHAPTER 6

ELEMENTS AND APPLICATIONS OF THE THEORY OF ANALYTIC FUNCTIONS

INTRODUCTION

The role played by the theory of analytic functions in physics has changed considerably over the past few decades. It no longer suffices to be able to work out residue integrals; a deeper understanding of the mathematical ideas has become essential if one wants to follow current applications to physical theory. Therefore the emphasis here will be on introducing the mathematical concepts and the logical structure of the theory of analytic functions. Assuming only that the reader is familiar with the properties of complex numbers, we aim to present a self-contained account of this theory in a way that prepares one to cope with modern applications of the theory as well as those of the past.

"Imaginary" numbers were discovered in the Middle Ages in the search for a general solution of quadratic equations. It is clear from the name given them that they were regarded with suspicion. Gauss, in his doctoral thesis of 1799, gave the now familiar geometrical representation of complex numbers, and thus helped to dispel some of the mystery about them. In this century, the trend has been toward defining complex numbers as abstract symbols subject to certain formal rules of manipulation. Thus complex numbers never have taken on the "earthy" qualities of real numbers. In fact, more nearly the opposite has occurred: we have come to view real numbers abstractly as symbols obeying their own set of axioms, just like complex numbers. We now speak of *number fields*: the real field and the complex field. The axioms which define a field were stated in Chapter 3 on vector spaces.

The theory of complex numbers can be developed by viewing them as ordered pairs of real numbers, written (x, y). Let (a, b) and (c, d) be two different complex numbers, and let K be a real number. Then we define addition, multiplication of a real and a complex number, and multiplication of two complex numbers by the following rules:

1. $(a, b) + (c, d) = (a + b, c + d)$,
2. $K \cdot (a, b) = (Ka, Kb)$,
3. $(a, b) \cdot (c, d) = (ac - bd, bc + ad)$.

From these definitions, we see that the set of all complex numbers—the complex plane—has the same mathematical structure as the set of all vectors in a plane.

This approach is followed in Landau's *Foundations of Analysis*, in which the various number systems are built up logically from Peano's five axioms; the

imaginary number i is never mentioned. However, if we write the ordered pair (a, b) as $a + ib$, where $i^2 = -1$, then the above rule of complex-number multiplication is obeyed if we simply multiply out the product $(a + ib)(c + id)$ according to the usual rules of multiplication of reals. The introduction of the symbol i subsumes the ordering aspect of the ordered pair of real numbers, while extending the formal rules of arithmetic from real to complex numbers.

From the complex numbers constructed as ordered pairs of reals, where $(a, b) = a + ib$, it is possible to generalize to hypercomplex numbers of three or more components, for example $(a, b, c) = a + ib + kc$. The four-component quaternions, a type of hypercomplex number which satisfies all the rules of arithmetic except the commutative law of multiplication, are useful in dealing with rotations of a rigid body. The four 4×4 Dirac matrices, $\gamma_i (i = 1, 2, 3, 4)$, form a set of hypercomplex numbers which satisfy the anticommutative relations

$$\gamma_i\gamma_j + \gamma_j\gamma_i = 2\delta_{ij} .$$

It can be shown that no matter how we define addition and multiplication for these hypercomplex numbers, it is impossible to retain all the usual rules of arithmetic. As Weyl points out, the complex numbers form a natural boundary for the extension of the number concept in this respect.

6.1 ANALYTIC FUNCTIONS—THE CAUCHY-RIEMANN CONDITIONS

If to each complex number z in a certain domain there corresponds another complex number w, then w is a function of the complex variable z: $w = f(z)$. If the correspondence is one to one, we can view this as a mapping from one plane (or part of it), the z-plane, to another, the w-plane. The complex functions thus defined are equivalent to ordered pairs of real functions of two variables, because w is a complex number depending on $z = x + iy$ and therefore can be written in the form

$$w(z) = u(x, y) + iv(x, y) .$$

However, this class of functions is too general for our purposes. We are interested only in functions which are differentiable with respect to the complex variable z—a restriction which is much stronger than the condition that u and v be differentiable with respect to x and y. Therefore one of our first tasks in the study of complex function theory will be to determine the necessary and sufficient conditions for a complex function to have a derivative with respect to the complex variable z. Single-valued functions of a complex variable which have derivatives throughout a region of the complex plane are called *analytic* functions. We shall restrict our attention to this special class of complex functions.

Two examples of complex functions (both written in the form $w = u + iv$) are

1. $w = z^* = x - iy$,
2. $w = z^2 = (x + iy)^2 = x^2 - y^2 + i2xy$.

Presently, we shall show that (1) is not an analytic function, but that (2) is analytic everywhere in the complex plane; i.e., its derivative exists at all points.

Before stating exactly what is meant by the derivative of a function of a complex variable, we must have a notion of *continuity* for these functions.

In the definition that follows, mention is made of the *absolute value* of a complex number, denoted by $|z|$. The reader will recall that $|z| \equiv (zz^*)^{1/2} = (x^2 + y^2)^{1/2}$. The absolute value is sometimes called the *modulus*.

Definition. A complex function $w = f(z)$ is continuous at the point z_0 if, given any $\epsilon > 0$, there exists a δ such that $|f(z) - f(z_0)| < \epsilon$, when $|z - z_0| < \delta$, or $f(z)$ is continuous at z_0 if

$$\lim_{z \to z_0} f(z) = f(z_0) .$$

This definition is formally exactly like the definition of continuity for real functions of a real variable. However, here the absolute value signs mean that whenever z lies within a *circle* of radius δ centered at z_0 in the complex z-plane, then $f(z)$ lies within a *circle* of radius ϵ centered at $f(z_0)$ in the complex w-plane. If $f(z) = u(x, y) + iv(x, y)$, then $f(z)$ is continuous at $z_0 = x_0 + iy_0$ if u and v are continuous at (x_0, y_0).

From the class of single-valued, continuous complex functions, we now want to select those that can be differentiated. Patterning the definition of a derivative after that of real analysis, we have

Definition. $f(z)$ is differentiable at the point z_0 if the limit

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{\Delta z \to 0} \frac{\Delta f}{\Delta z}$$

exists. We shall denote this limit, the derivative of $f(z)$ at z_0, by $f'(z_0)$.

A very important feature of the limits that occur in the definitions of continuity and the derivative is that z may approach z_0 from *any direction* on the plane. When we say the limit exists, we therefore mean that the same number must result from the limiting process regardless of how the limit is taken. This is also true in real analysis, but in that case there are only two possible directions of approach in taking the limit: from the left or the right on the real line. In real analysis, the limiting process is one-dimensional; in complex analysis, it is two-dimensional.

The equation that defines the derivative means that given any $\epsilon > 0$, there exists a δ such that

$$\left| f'(z) - \frac{f(z) - f(z_0)}{z - z_0} \right| < \epsilon$$

provided $|z - z_0| < \delta$. The requirement that the ratio $[f(z) - f(z_0)]/(z - z_0)$ always tends to the same limiting value, no matter along what path z approaches z_0, is an extremely exacting condition. The theory of analytic functions contains a number of amazing theorems, and they all result from this stringent initial requirement that the functions possess "isotropic" derivatives.

A single-valued function of z is said to be *analytic* (or *regular*) at a point z_0 if it has a derivative at z_0 and at all points in some neighborhood z_0. Thus a slight distinction is drawn between differentiability and analyticity. It pays to do this, because although there exist functions which have derivatives at certain points, or even along certain curves, no interesting results can be obtained unless functions are differentiable throughout a region, i.e., unless they are analytic. Thus if we say a function is analytic on a curve, we mean that it has a derivative at all points in a two-dimensional strip containing the curve. If a function is not analytic at a point or on a curve, we say it is *singular* there.

We shall now examine the two complex functions mentioned earlier for differentiability and analyticity. We write the derivative at z_0 in the form

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z},$$

by letting $z = z_0 + \Delta z$ in the original definition. For $f(z) = z^2$, we have

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{(z_0 + \Delta z)^2 - z_0^2}{\Delta z} = \lim_{\Delta z \to 0} (2z_0 + \Delta z) = 2z_0,$$

a result which is clearly independent of the path along which $\Delta z \to 0$, so $f(z) = z^2$ is differentiable and analytic everywhere. The result parallels exactly the result for the derivative of the real function $f(x) = x^2$.

On the other hand, if $f(z) = z^*$, we have

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{z_0^* + \Delta z^* - z_0^*}{\Delta z} = \lim_{\Delta z \to 0} \frac{\Delta z^*}{\Delta z}.$$

Now if $\Delta z \to 0$ along the real x-axis, then $\Delta z = \Delta x$ and $\Delta z^* = \Delta x^* = \Delta x$, so $f'(z_0) = +1$. However, if Δz approaches zero along the imaginary y-axis, then $\Delta z = i\Delta y$ so $\Delta z^* = -i\Delta y = -\Delta z$, so $f'(z_0) = -1$. Since at any point z_0 the limit as $z \to z_0$ depends on the direction of approach, the function is not differentiable or analytic anywhere. [As a general rule, $\Delta z^*/\Delta z = e^{-2i\theta}$, where $\theta = \tan^{-1}(\Delta y/\Delta x)$, which manifestly involves the direction of approach (θ) in taking the limit.]

Many of the theorems on differentiability in real analysis have analogs in complex analysis. For example:

1. A constant function is analytic.
2. $f(z) = z^n$ $(n = 1, 2, \cdots)$ is analytic.
3. The sum, product, or quotient of two analytic functions is analytic, provided, in the case of the quotient, that the denominator does not vanish anywhere in the region under consideration.
4. An analytic function of an analytic function is analytic.

The proofs go through exactly as in the real case.

We now determine the necessary and sufficient conditions for a function $w(z) = u(x, y) + iv(x, y)$ to be differentiable at a point. First, we assume that

$w(z)$ is in fact differentiable for some $z = z_0$. Then

$$w'(z_0) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \to 0} \left(\frac{\Delta u}{\Delta z} + i \frac{\Delta v}{\Delta z} \right).$$

Since $w'(z_0)$ exists, it is independent of how $\Delta z \to 0$; that is, it is independent of the ratio $\Delta y / \Delta x$. If the limit is taken along the real axis, $\Delta y = 0$, and $\Delta z = \Delta x$. Then

$$w'(z_0) = \lim_{\Delta x \to 0} \left(\frac{\Delta u}{\Delta x} + i \frac{\Delta v}{\Delta x} \right) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}.$$

On the other hand, if we approach the origin along the imaginary axis, $\Delta x = 0$ and $\Delta z = i \Delta y$. Now

$$w'(z_0) = \lim_{\Delta y \to 0} \left(\frac{\Delta v}{\Delta y} - i \frac{\Delta u}{\Delta y} \right) = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}.$$

But by the assumption of differentiability, these two limits must be equal. Therefore, equating real and imaginary parts, we have

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}. \tag{6.1}$$

Equations (6.1) are known as the Cauchy-Riemann equations. They give a *necessary* condition for differentiability. We have determined this condition from special cases of the requirement of differentiability; therefore it is not surprising that these conditions alone are not sufficient.

The sufficient conditions for the differentiability of $w(z)$ at z_0 are, first, that the Cauchy-Riemann equations hold there, and second, that the first partial derivatives of $u(x, y)$ and $v(x, y)$ exist and be continuous at z_0.

The proof is straightforward. To begin, u is continuous at (x_0, y_0) because it is differentiable there; the partial derivatives of u are continuous by hypothesis. Under these assumptions, it follows from the calculus of functions of several variables* that

$$\begin{aligned} \Delta u &= u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0) \\ &= \frac{\partial u}{\partial x} \Delta x + \frac{\partial u}{\partial y} \Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y, \end{aligned}$$

where $\partial u / \partial x$ and $\partial u / \partial y$ are the partial derivatives evaluated at the point (x_0, y_0) and where ϵ_1 and ϵ_2 go to zero as both Δx and Δy go to zero. Using a similar formula for $v(x, y)$, we have

$$\begin{aligned} \Delta w &= w(z_0 + \Delta z) - w(z_0) = \Delta u + i \Delta v \\ &= \frac{\partial u}{\partial x} \Delta x + \frac{\partial u}{\partial y} \Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y + i \left(\frac{\partial v}{\partial x} \Delta x + \frac{\partial v}{\partial y} \Delta y + \epsilon_3 \Delta x + \epsilon_4 \Delta y \right). \end{aligned}$$

* See, for example, G. B. Thomas, Jr. *Calculus and Analytic Geometry*, 4th Ed., Addison-Wesley Publishing Co., 1968, Section 15-4, p. 503 Eq. 4, or W. Kaplan, *Advanced Calculus*, Addison-Wesley Publishing Co., 1953, Section 2-6, p. 84.

Now using the Cauchy-Riemann equations, which by assumption hold at the point (x_0, y_0), we have

$$\Delta w = \frac{\partial u}{\partial x}(\Delta x + i\Delta y) + i\frac{\partial v}{\partial x}(\Delta x + i\Delta y) + \Delta x(\epsilon_1 + i\epsilon_3) + \Delta y(\epsilon_2 + i\epsilon_4) .$$

Therefore

$$\frac{\Delta w}{\Delta z} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} + (\epsilon_1 + i\epsilon_3)\frac{\Delta x}{\Delta z} + (\epsilon_2 + i\epsilon_4)\frac{\Delta y}{\Delta z} .$$

Since $|\Delta z| = [(\Delta x)^2 + (\Delta y)^2]^{1/2}$, $|\Delta x| \leq |\Delta z|$ and $|\Delta y| \leq |\Delta z|$, and so $|\Delta x/\Delta z| \leq 1$ and $|\Delta y/\Delta z| \leq 1$. Since these factors are bounded, the last two terms in the above equation tend to zero with Δz because ϵ_1, ϵ_2, ϵ_3, and ϵ_4 go to zero as Δz goes to zero. Therefore at z_0

$$w'(z_0) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} ; \tag{6.2}$$

the limit is independent of the path followed, so the derivative exists. Using the Cauchy-Riemann conditions, we also have

$$w'(z_0) = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y} . \tag{6.3}$$

Example. Consider the function z^3. We have

$$z^3 = (x^3 - 3xy^2) + i(3x^2y - y^3) = u + iv .$$

Thus

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 = \frac{\partial v}{\partial y}, \qquad \text{and} \qquad \frac{\partial v}{\partial x} = 6xy = -\frac{\partial u}{\partial y} .$$

Thus the Cauchy-Riemann equations hold everywhere. Since the partial derivatives are continuous, the function z^3 is, in fact, analytic everywhere. A function which is analytic in the entire complex plane is said to be an *entire* function. The derivative of z^3 may be found using Eq. (6.2) or (6.3). We obtain

$$\frac{\partial z^3}{\partial z} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = 3[(x^2 - y^2) + 2ixy] = 3z^2 ,$$

a satisfying result. As a second example, we leave it to the reader to show that the function $|z|^2 \equiv zz^*$ is differentiable only at the origin, and therefore is analytic nowhere.

One remarkable result which points to connections with physics follows immediately from the Cauchy-Riemann equations. Assuming that they hold in a region, we have

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x\,\partial y} = \frac{\partial^2 v}{\partial y\,\partial x} = -\frac{\partial^2 u}{\partial y^2} \Longrightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \nabla^2 u = 0 \tag{6.4}$$

if the second partial derivatives are continuous, so we can interchange the orders

of differentiation in the mixed partial derivative. It follows in the same way that the function v also satisfies the two-dimensional Laplace equation. Thus both the real and imaginary parts of an analytic function with continuous second partial derivatives satisfy the two-dimensional Laplace equation. We shall later prove, using integration theory, that the second partial derivatives of an analytic function are necessarily continuous, so this qualification can be dropped. (It is interesting that these theorems about derivatives can be proved only by integration.) Any function ϕ satisfying $\nabla^2\phi = 0$ is called a *harmonic function*. If $f = u + iv$ is an analytic function, then $\nabla^2 u = \nabla^2 v = 0$, and u and v are called *conjugate* harmonic functions.

Given one of two conjugate harmonic functions, the Cauchy-Riemann equations can be used to find the other, up to a constant. For example, the function $u(x, y) = 2x - x^3 + 3xy^2$ is easily seen to be harmonic. To find its harmonic conjugate, we proceed as follows:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 2 - 3x^2 + 3y^2 \Longrightarrow v = 2y - 3x^2y + y^3 + \phi(x) ,$$

where $\phi(x)$ is some function of x. Now, using the other Cauchy-Riemann equation, we obtain

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \Longrightarrow -6xy + \phi'(x) = -6xy \Longrightarrow \phi' = 0 .$$

Thus $\phi(x)$ must be a constant, and the harmonic conjugate of u is

$$v = 2y - 3x^2y + y^3 + \text{const.}$$

Note that the function $w = u + iv = 2z - z^3 + C$ is an analytic function, as we know it must be.

Before leaving the Cauchy-Riemann conditions, let us take advantage of being physicists to present another, shorter derivation of these conditions, based on the use of infinitesimals. Let $w = u + iv$ and $w' = p + iq$. Then $\delta w = w'\delta z$, or, taking real and imaginary parts,

$$\delta u = p\delta x - q\delta y , \qquad \delta v = p\delta y + q\delta x .$$

It follows immediately that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = p , \qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = q .$$

These equations are identical to the Cauchy-Riemann equations (6.1).

Continuing in this informal spirit, we may derive another closely related result which provides some insight into the meaning of analyticity. Again, let $w(z) = w(x, y) = u(x, y) + iv(x, y)$. We now show that $\partial w/\partial z^* = 0$ if and only if the Cauchy-Riemann equations hold. We shall not worry about the meaning of this derivative with respect to z^*, but just differentiate formally, treating the

derivative as symbolic. Using the expressions

$$x = (z + z^*)/2 \qquad \text{and} \qquad y = (z - z^*)/2i$$

we have

$$\begin{aligned}\frac{\partial w}{\partial z^*} &= \frac{\partial w}{\partial x}\frac{\partial x}{\partial z^*} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial z^*} \\ &= \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right)\tfrac{1}{2} + \left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right)\left(-\frac{1}{2i}\right) \\ &= \tfrac{1}{2}\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) + \frac{i}{2}\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right).\end{aligned}$$

If the Cauchy-Riemann equations hold, this last expression vanishes. If, on the other hand, $\partial w/\partial z^* = 0$, then both the real and imaginary parts of the last expression must vanish, so the Cauchy-Riemann equations hold.

This purely formal result, which can be made rigorous, is trying to tell us that analytic functions are independent of z^*: they are functions of z *alone*. Thus analytic functions are true functions of a *complex* variable, not just complex functions of two real variables (see, for example, Problem 1), which will in general depend on z^* as well as z according to

$$f(x, y) = f\left(\frac{z + z^*}{2}, \frac{z - z^*}{2i}\right).$$

6.2 SOME BASIC ANALYTIC FUNCTIONS

One of the most useful functions in the complex domain is the exponential function which we define for $z = x + iy$ by

$$e^z \equiv e^x (\cos y + i \sin y) . \tag{6.5}$$

It follows easily from this definition and our earlier work that e^z is an entire function and that

$$\frac{d}{dz} e^z = e^z .$$

The other familiar properties of exponentials, in particular, $e^{z_1+z_2} = e^{z_1} e^{z_2}$, follow readily from Eq. (6.5). We note that e^z is a periodic function of period $2\pi i$:

$$e^{z+2\pi i} = e^z e^{2\pi i} = e^z (\cos 2\pi + i \sin 2\pi) = e^z .$$

From Eq. (6.5) we see that

$$e^{iy} = \cos y + i \sin y ,$$

so it follows that

$$\cos y = \frac{e^{iy} + e^{-iy}}{2}, \qquad \sin y = \frac{e^{iy} - e^{-iy}}{2i}.$$

These relations suggest that for an arbitrary complex z we define

$$\cos z \equiv \frac{e^{iz} + e^{-iz}}{2}, \tag{6.6}$$

$$\sin z \equiv \frac{e^{iz} - e^{-iz}}{2i}. \tag{6.7}$$

Since

$$\frac{d}{dz} e^z = e^z,$$

it is a simple matter to calculate the derivatives of $\cos z$ and $\sin z$. We find that

$$\frac{d}{dz} \cos z = \frac{ie^{iz} - ie^{-iz}}{2} = -\sin z,$$

$$\frac{d}{dz} \sin z = \frac{ie^{iz} + ie^{-iz}}{2i} = \cos z,$$

as we might expect from experience with the real variable case. Using Eqs. (6.6) and (6.7), it is a simple matter to verify that all the familiar trigonometric identities, such as

$$\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2,$$

continue to be valid for complex variables.

The complex functions sine and cosine may, of course, be put in the form $u(x, y) + iv(x, y)$. For example,

$$\begin{aligned}\sin z &= \frac{1}{2i}[e^{i(x+iy)} - e^{-i(x+iy)}] \\ &= \frac{1}{2i} e^{-y}(\cos x + i \sin x) - \frac{1}{2i} e^{y}(\cos x - i \sin x) \\ &= \sin x(e^y + e^{-y})/2 + i \cos x(e^y - e^{-y})/2.\end{aligned}$$

Therefore

$$\sin z = \cosh y \sin x + i \sinh y \cos x. \tag{6.8}$$

Similarly,

$$\cos z = \cosh y \cos x - i \sinh y \sin x. \tag{6.9}$$

Setting $x = 0$, we obtain the useful relations $\sin(iy) = i \sinh y$ and $\cos(iy) = \cosh y$. We also see that the Cauchy-Riemann conditions are satisfied everywhere, as we know they must be. Other properties which follow directly from Eqs. (6.8) and (6.9) are

$$(\sin z)^* = \sin(z^*),$$

$$\sin(-z) = -\sin(z),$$

$$\sin(z + 2\pi) = \sin(z).$$

Using the sine and cosine functions, we can define the other familiar trigonometric functions. For example,

$$\tan z = \sin z/\cos z \ ;$$

similar extensions of the real case are defined for the cotangent, secant, and cosecant. These functions differ from the sine and cosine in that they are *not* analytic everywhere. The tangent, being the ratio of two analytic functions, will be analytic everywhere *except* at points where $\cos z = 0$. Using the real and imaginary parts of the cosine, we can rewrite this condition as

$$\cosh y \cos x = 0 \ , \qquad \sinh y \sin x = 0 \ .$$

Now $\cosh y \geq 1$ for all real y, so the first equation has a solution whenever $\cos x = 0$, or $x = (2n + 1)\pi/2$, $n = 0, \pm 1, \pm 2, \cdots$. At these points, $\sin x = \pm 1$, so the second equation requires that $\sinh y = 0$, that is, $y = 0$. Thus the tangent function is singular at the points $(2n + 1)\pi/2$, $(n = 0, \pm 1, \cdots)$ on the real axis, and only at these points. Therefore $\tan z$ becomes infinite at precisely those points where $\tan x$ (real x) becomes infinite and *only* at those points.

On the basis of the above discussion, one might be tempted to think that the complex trigonometric functions are "just the same thing" as their real counterparts. However, the reader can easily show that

$$|\sin z|^2 = \sin^2 x + \sinh^2 y \ ,$$

and this expression increases *without limit* as y tends to infinity. This is in marked contrast with the real case, where $|\sin x| \leq 1$ for all real x.

The functions which we have discussed thus far all have the property that if we pick any point z_0 in the complex plane and follow any path from z_0 through the plane back to z_0, then the value of the function changes continuously along the path, returning to its original value at z_0. For example, suppose that we consider the function $w(z) = e^z$ and start at the point $z_0 = 1$, encircling the origin in the z-plane counterclockwise along the unit circle. Figure 6.1(a) shows the circular path in the z-plane, and Fig. 6.1(b) shows the corresponding path in the w-plane. [The use of two complex planes to "graph" the function $w(z)$ is often employed in complex variable theory.] We note that both paths are closed, which is just the geometrical statement of the fact that if we start at a point z_0, where the function has the value $w(z_0)$, then when we move along a closed curve back to z_0, the functional values also follow a smooth path back to $w(z_0)$.

Now for e^z this result is hardly surprising since we have *defined* e^z in such a way as to ensure this behavior, letting ourselves be guided by the properties of the real exponential function. Now if we look at another simple function, namely, the square root, we see that things do not always go so smoothly. Let us write formally

$$w(z) \equiv \sqrt{z} \equiv \sqrt{x + iy} \ .$$

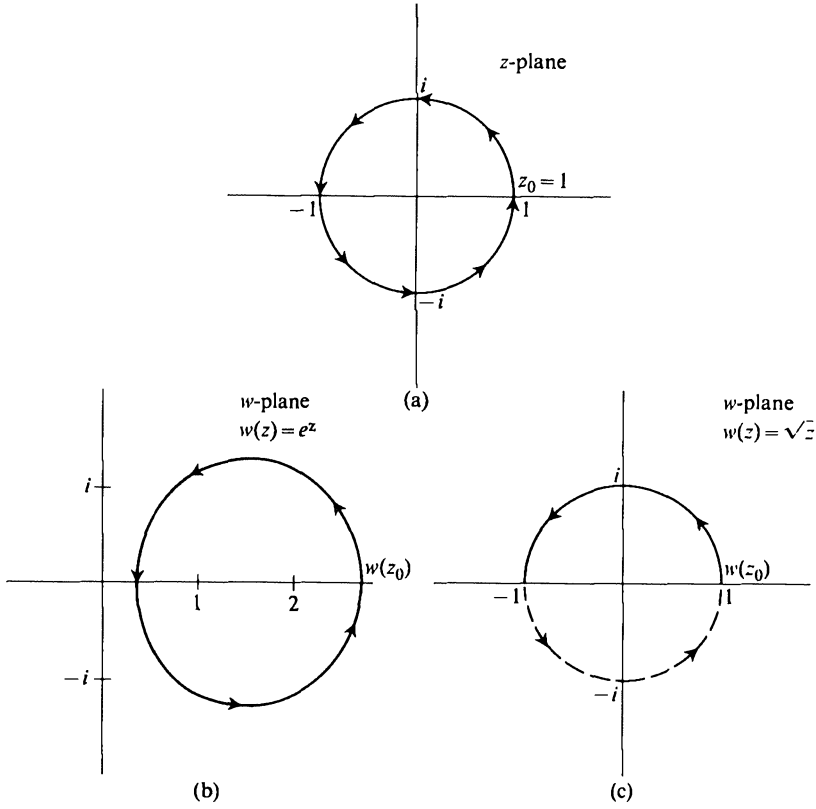

Fig. 6.1(a) A circular contour in the z-plane about the origin. **6.1(b)** The mapping of the contour of Figure 6.1(a) by the function e^z. **Fig. 1(c)** The mapping of the contour of Figure 6.1(a) by the function $\sqrt{z}$.

We observe that this definition is empty, since there is no set of operations presently at our disposal which will enable us to find $w(z)$ for some given x and y (unless $y = 0$). This is in contrast with the situation in Eq. (6.5) where all problems of evaluation can be handled by familiar real-variable operations.

Fortunately, in the case of the square root there is another possibility, namely, we can write z in polar form† as $z = re^{i\theta}$. In this form, a logical extension of the square root to the complex domain is contained in the definition

$$w(z) \equiv \sqrt{z} \equiv \sqrt{r}\, e^{i\theta/2} = \sqrt{r}\,[\cos(\theta/2) + i \sin(\theta/2)]\,.$$

† If we write $z = x + iy$ and make the familiar change to polar coordinates ($x = r\cos\theta$, $y = r\sin\theta$) we obtain $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$, where $r = (x^2 + y^2)^{1/2}$ and $\theta = \tan^{-1}(y/x)$.

Clearly, this function satisfies $w^2 = z$, which is certainly a minimum requirement for any sensible square root. Using this definition, let us vary z along the same path chosen in Fig. 6.1(a), starting at $r = 1, \theta = 0$. Figure 6.1(c) shows the corresponding path in the w-plane. Note that it is *not* a closed path; after making a complete circle around the origin in the z-plane, we arrive at the point $w = -1$ in the w-plane, not at $w = +1$. In order to get back to $w = +1$, we must let θ go from 2π to 4π; that is, make the circular trip in the z-plane one more time [see the dotted curve in Fig. 6.1(c)]. Actually this is not quite the best way to describe the situation; we do not want to think of tracing the circular path in the original z-plane a second time, but rather of tracing an identical circular path in a *different* z-plane. This corresponds to the fact that in the first circuit, θ went from 0 to 2π, whereas in the second circuit, it went from 2π to 4π.

This is not so different from the case of functions like e^z as one might imagine at first glance. We can write in polar variables

$$e^z = e^{r\cos\theta}\,[\cos(r\sin\theta) + i\sin(r\sin\theta)]\,,$$

and then trace out a circular path as many times as we please ($\theta = 0 \to \theta = 2\pi$, $\theta = 2\pi \to \theta = 4\pi$, etc.). In this case we get the same values of e^z for each circuit. Therefore no information about e^z is lost if we identify the z-planes corresponding to $\theta = 0 \to \theta = 2\pi$, $\theta = 2\pi \to \theta = 4\pi$, etc., with each other. However, in the case of $w(z) = \sqrt{z}$ we need two planes, usually referred to as *Riemann sheets*, to characterize the values of $w(z)$ in a single-valued manner. Two planes are clearly sufficient: when we let θ range from 4π to 6π we obtain the same values as we did when we let θ range from 0 to 2π.

It is important to remember that the path of Fig. 6.1(a) encloses the origin. If we choose a closed path which neither encloses the origin nor intersects the positive real axis, then we also obtain a closed path in the w-plane. Fig. 6.2(a) and (b) illustrates the situation for $w(z) = \sqrt{z}$, starting from the $z_0 = \sqrt{5}\,e^{i\phi_0}$, where $\phi_0 = \tan^{-1} 2$ (we adopt the usual trigonometric convention that $\tan^{-1} x$ takes on values between 0 and $\pi/2$). In Fig. 6.2(a) we may say that we start at $z_0 = 1 + 2i$ on the *first Riemann sheet* and return to that point without encircling the origin. If we do the same thing for $z_0 = 1 + 2i$ on the *second Riemann sheet* (that is, $\phi_0 = \tan^{-1} 2 + 2\pi$), we obtain the corresponding closed curve traced out in the w-plane (Fig. 6.2(c)).

It is readily seen that the difficulties described above for $w(z) = \sqrt{z}$ will persist for any path beginning on the positive real axis and returning to the original point along a path enclosing the origin. Thus if we wish to consider $w(z) = \sqrt{z}$ in the simple fashion that we used for e^z, then we conclude that $\sqrt{z}$ is not continuous along the positive real axis and is not analytic there. However, to avoid this dilemma, we can say that when we come back to the real axis after a circuit of 2π radians, we transfer continuously onto the *second* Riemann sheet. If we go around $z = 0$ once more on the second sheet, when we return toward the positive real axis, we transfer continuously back to the *first* Riemann sheet. Thus the two sheets can be imagined to be *cut* along the positive real axis and

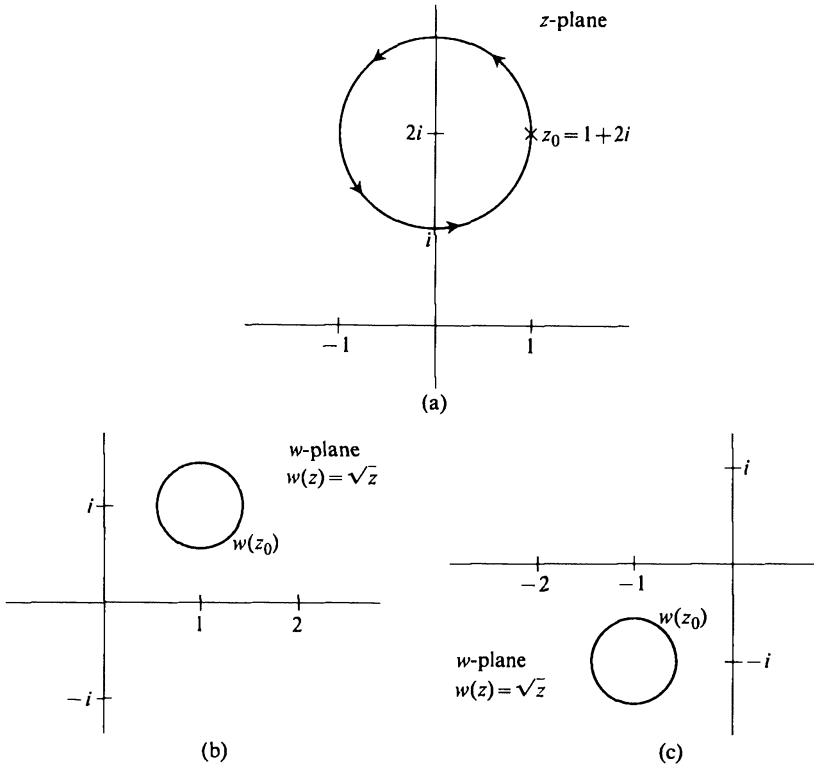

Fig. 6.2(a) A closed contour in the z-plane which does not enclose the origin. **6.2(b)** The mapping of the contour of Figure 6.2(a) by the function $\sqrt{z}$, if the contour of Figure 6.2(a) is imagined to lie on the *first* Riemann sheet. **6.2(c)** The mapping of the contour of Figure 6.2(a) by the function $\sqrt{z}$, if the contour of Figure 6.2(a) is imagined to lie on the *second* Riemann sheet.

joined in the manner illustrated in Fig. 6.3. With this construction, the function $w(z) = \sqrt{z}$ is seen to be single valued everywhere [on both sheets we set $w(0) = 0$] and analytic everywhere except at the origin, where $\sqrt{z}$ suffers from the same difficulty as does $\sqrt{x}$ in the real variable case. Thus the origin is a singular point for $w(z) = \sqrt{z}$.

In general, suppose that we have a singular point, z_0, of some function $w(z)$, and a path starting at z_1, which encircles z_0. If we must sweep through an angle

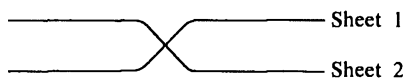

Fig. 6.3 A side view of the two Riemann sheets, looking down the real axis towards the origin, for the function $w(z) = \sqrt{z}$.

greater than 2π in order to return to the original value at z_1, then z_0 is called a *branch point* of $w(z)$. The cut which emanates from this point is called a *branch cut*. In the simple case discussed above $[w(z) = \sqrt{z}]$, the value of $w(z_1)$ with z_1 on the *second* sheet is just the negative of $w(z_1)$ with z_1 on the *first* sheet. Note that it is possible for a point z_1 on the *first* sheet to be a point at which some function $w(z)$ is analytic, whereas the point z_1 on the *second* sheet is a singular point. The function

$$w(z) = \frac{1}{i + \sqrt{z}}$$

is an example of such a function; $w(z)$ is a single-valued function over a two-sheeted Riemann surface cut along the positive real axis and joined as in Fig. 6.3. It is analytic at $z = -1\,(\phi = \pi)$ on the first Riemann sheet and is singular at $z = -1\,(\phi = 3\pi)$ on the second Riemann sheet.

In the above discussions we could, of course, have insisted that θ, the argument of z, range only through 2π radians. Then we could say that

$$w_1(z) = \sqrt{r}\, e^{i\theta/2}, \qquad 0 \le \theta < 2\pi,$$

defines a single-valued function, analytic everywhere in the complex z-plane *except along the positive real axis* (including $z = 0$), and that

$$w_2(z) = \sqrt{r}\, e^{i(\theta+2\pi)/2} = -\sqrt{r}\, e^{i\theta/2}, \qquad 0 \le \theta < 2\pi,$$

also defines such a function. Both $w_1(z)$ and $w_2(z)$ satisfy $w_1^2 = w_2^2 = z$ and are referred to as *single-valued branches* of $\sqrt{z}$. Clearly, if we defined

$$w_3(z) = \sqrt{r}\, e^{i(\theta+4\pi)/2}, \qquad 0 \le \theta < 2\pi,$$

we would find $w_3(z) = w_1(z)$, so we do not obtain a new branch in this manner. We leave it as an exercise to show that one can define *three* single-valued branches of $w(z) = \sqrt[3]{z}$ and that, in this case, if one wants to define $w(z)$ as a single-valued function, analytic everywhere except at $z = 0$, a three-sheeted Riemann surface is necessary.

It should be noted that the choice of the real axis as the branch cut for $w(z) = \sqrt{z}$ was entirely arbitrary. Any other ray, say $\theta = \theta_0$, will serve equally well. The only thing which is *not* arbitrary is the choice of $z = 0$ as a branch point; $z = 0$ is a bona fide singular point for $w(z) = \sqrt{z}$, and this cannot be changed. However, in the functions $w_1(z)$ and $w_2(z)$ defined above, the singular line $\theta = 0$ is, apart from $z = 0$, a line of "man-made" singularities; we could equally well choose the line $\theta = \theta_0$ to be the singular line. For example, we could define

$$w_1(z) = \sqrt{r}\, e^{i\theta/2}, \qquad \theta_0 \le \theta < \theta_0 + 2\pi,$$

and, similarly,

$$w_2(z) = \sqrt{r}\, e^{i(\theta+2\pi)/2} = -\sqrt{r}\, e^{i\theta/2}, \qquad \theta_0 \le \theta < \theta_0 + 2\pi.$$

If we want a single-valued function which is analytic everywhere except at $z = 0$,

then we can construct a two-sheeted Riemann surface, cut and joined along the line $\theta = \theta_0$. On the counterclockwise edge of the cut, $\sqrt{z} = \sqrt{r}\, e^{i\theta_0/2}$. After going through a circuit of 2π radians, we do not return to this value, but to $-\sqrt{r}\, e^{i\theta_0/2}$, and we pass onto the second sheet. Note that, in this case, some of the values of $w(z) = \sqrt{z}$ which were on the *first* Riemann sheet when the cut was made along $\theta = 0$ now find themselves on the *second* Riemann sheet and vice versa. This brings home the fact that the Riemann construction is merely a way to write a collection of values of a function in a single-valued manner. Distinctions between the first sheet and the second sheet are purely matters of convention. In fact, it should not be difficult for the reader to imagine that any reasonable curve from the origin to infinity could serve as an acceptable cut along which the two Riemann sheets of $w(z) = \sqrt{z}$ can be joined.

Before leaving this example, let us first propose an argument which might appear at first sight to contradict what we have been saying. Consider any point $x_0 \neq 0$ on the positive real axis. We should imagine that there exists a neighborhood of x_0 in which we can write $z = x_0 + \rho$, where ρ is some complex number, and then define $\sqrt{z}$ by the power series

$$\sqrt{z} \equiv \sqrt{x_0}\left(1 + \tfrac{1}{2}\frac{\rho}{x_0} - \tfrac{1}{8}\frac{\rho^2}{x_0^2} + \cdots\right),$$

whenever the series converges (we will see later that the series converges whenever $|\rho| < x_0$). This is a single-valued function which defines $\sqrt{z}$ continuously across a part of the positive real axis. However, this definition does not apply to the whole complex plane since the series does not converge everywhere. It turns out that it is impossible to extend (or "continue") this function to all points of the z-plane in such a way that $\sqrt{z}$ is single-valued and analytic. We will return to this point when we discuss the principle of analytic continuation later in this chapter. The continuity of the series definition across the positive real axis does not contradict our original positioning of a cut along the positive real axis, because, as we have seen above, the cut could be positioned anywhere so long as it begins at $z = 0$. In particular, the cut can be chosen so that it lies completely outside the domain of convergence of the series used to define $\sqrt{z}$ (for example, the cut could be chosen to lie along the negative real axis). We may remark in passing that we *could* successfully define e^z in this manner. The function e^x possesses an everywhere convergent power series for real x, and it is not hard to believe that a complex power series with the same coefficients $(1/n!)$ will converge *everywhere* in the complex plane.

As another example of a multivalued function, we consider the logarithm. Again using $z = re^{i\theta}$, we define

$$\log z \equiv \ln r + i\theta\,,$$

where ln denotes the usual natural logarithm of a positive real number. Note that

$$e^{\log z} = e^{\ln r + i\theta} = e^{\ln r} e^{i\theta} = re^{i\theta} = z$$

and also that

$$\log(z_1 z_2) = \ln(r_1 r_2) + i(\theta_1 + \theta_2) = \ln r_1 + i\theta_1 + \ln r_2 + i\theta_2 = \log z_1 + \log z_2 ,$$

so the logarithm has the main properties that one would expect by analogy with the real variable case. With the logarithm, the multivaluedness difficulties described above are more striking, since no matter how many times one encircles the origin, starting, say, at some point on the positive real axis, one *never* returns to the original value of the logarithm. The logarithm increases by $2\pi i$ on each circuit (or decreases by $2\pi i$ if one moves in the direction of decreasing θ). Thus an infinite number of Riemann sheets, each one joined to the one below it via a cut along the positive real axis, is necessary to turn $\log z$ into a single-valued function. When this is done, $\log z$ is analytic everywhere except at $z = 0$, where we assign the value $\log(z = 0) = -\infty$ on all sheets. We can also form an infinite number of single-valued branches of the logarithm:

$$w_n(z) = \ln r + i\theta + 2\pi n i , \qquad 0 \leq \theta < 2\pi ,$$

where $n = 0, \pm 1, \pm 2, \cdots$, and $w_n(z)$ is a single-valued function, analytic everywhere except at $z = 0$ and along the positive real axis. Just as before,

$$e^{w_n(z)} = z ,$$

but

$$w_n(z_1 z_2) = w_n(z_1) + w_n(z_2) - 2\pi n i .$$

The branch

$$w_0(z) = \ln r + i\theta , \qquad 0 \leq \theta < 2\pi ,$$

is called the principal value or principal branch of the logarithm and is usually denoted by Log z. We have

$$e^{\operatorname{Log} z} = z ,$$

$$\operatorname{Log}(z_1 z_2) = \operatorname{Log} z_1 + \operatorname{Log} z_2 .$$

As before, the choice of the ray $\theta = 0$ as the line of singularities is entirely arbitrary.

From the preceding examples, it is a simple matter to build up to more complicated cases. For example, the function

$$w(z) = \sqrt{z - a}$$

is a single-valued function on the two-sheeted Riemann surface cut from a to infinity. The point $z = a$ is a branch point singularity; if we choose the branch cut to lie parallel to the real axis, we obtain the picture of Fig. 6.4. We thus may define, using the notation of Fig. 6.4,

$$w(z) \equiv |z - a|^{1/2} e^{i\theta/2} .$$

A more challenging problem is provided by the function

$$w(z) = \sqrt{(z - a)(z - b)} .$$

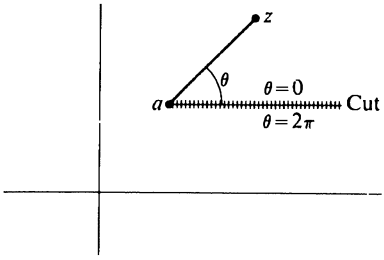

Fig. 6.4 The appropriate cut for the function $w(z) = \sqrt{z-a}$.

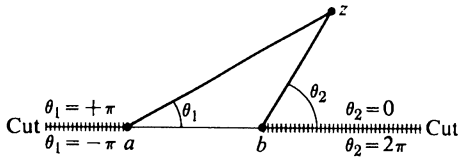

Fig. 6.5 The appropriate cuts for the function $w(z) = \sqrt{(z-a)(z-b)}$.

To simplify the geometry, we will consider the special case when a and b are real. The most obvious way to proceed is simply to put in two cuts from the branch points $z = a$ and $z = b$ as shown in Fig. 6.5. We have chosen these two simple directions for the cuts because of pictorial convenience. The left-hand cut corresponds to what we would have for the function $\sqrt{z-a}$, the right-hand cut to what we would have for $\sqrt{z-b}$. Here we take θ_1 to begin on the "bottom" of the left-hand cut at $-\pi$ and go to $+\pi$ on the "top" of the cut. At π we transfer to the second sheet and θ_1 continues to 3π, where we return to the first sheet. As we look down the negative real axis toward the origin, the two sheets are joined as shown in Fig. 6.3. Similarly, for the right-hand cut, we start at the "top" of the cut at $\theta_2 = 0$ and move counterclockwise, passing to the second sheet at $\theta_2 = 2\pi$, and finally return to the first sheet when $\theta_2 = 4\pi$. Along the positive real axis, the two sheets are joined as shown in Fig. 6.3. Using the above conventions, we define

$$w(z) \equiv |z-a|^{1/2}|z-b|^{1/2}e^{i\theta_1/2}e^{i\theta_2/2}.$$

As shown in Fig. 6.6, any point z_0 on the second sheet can be reached from the point z_0 on the first sheet either by going via the left-hand cut or the right-hand cut. These two options differ only in the sense that in the first case θ_1 increases by 2π, whereas in the second case θ_2 increases by 2π. In both cases, the value

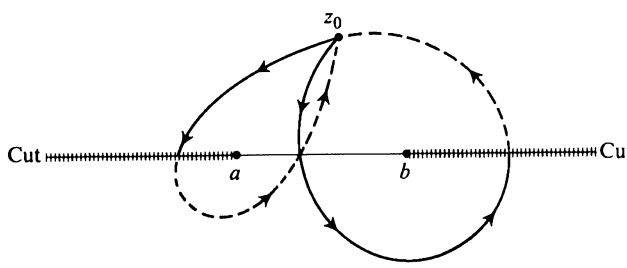

Fig. 6.6 Two different paths by which one can go from $z = z_0$ on the first sheet to $z = z_0$ on the second sheet. The function in question is $w(z) = \sqrt{(z-a)(z-b)}$.

of $w(z)$ is the same (the negative of what it would be for the corresponding point on the first sheet), as it must be if we are to have a single-valued function. With this construction, $w(z) = \sqrt{(z-a)(z-b)}$ becomes a single-valued function, analytic everywhere except at $z = a$ and $z = b$.

Note that in the above example, if we go through 2π radians in both θ_1 and θ_2, starting for example on the "top" of the right-hand cut, then we go down to the second sheet via the left-hand cut and return to our starting point on the first sheet via the right-hand cut. This suggests that it should equally well be possible to cut and join the two sheets along the real axis from a to b. The reader may find it interesting to show that this is indeed the case.

Using the above ideas, we can also obtain sensible expressions for the inverse trigonometric functions. For example, consider

$$w = \tan^{-1} z .$$

Writing this as

$$z = \tan w = \frac{1}{i}\frac{e^{iw} - e^{-iw}}{e^{iw} + e^{-iw}},$$

we obtain readily

$$(1 - iz)e^{iw} = (1 + iz)e^{-iw} ,$$

and hence

$$e^{2iw} = \frac{1 + iz}{1 - iz} .$$

Taking the logarithms of both sides, we find that

$$w = \frac{1}{2i}\log\left(\frac{1 + iz}{1 - iz}\right) = \frac{1}{2i}\log\left(\frac{i - z}{i + z}\right).$$

Thus

$$\tan^{-1} z = \frac{i}{2}\log\left(\frac{i + z}{i - z}\right).$$

Just as we did in discussing the logarithm, we can speak of single-valued branches of $\tan^{-1} z$ corresponding to single-valued branches of $\log z$. The principal branch of the inverse tangent is defined in the obvious manner:

$$\mathrm{Tan}^{-1} z = \frac{i}{2}[\mathrm{Log}\,(i + z) - \mathrm{Log}\,(i - z)] .$$

Similar considerations apply to the functions $\sin^{-1} z$ and $\cos^{-1} z$.

6.3 COMPLEX INTEGRATION—THE CAUCHY-GOURSAT THEOREM

We now come to integration of complex functions, the part of the theory that makes the subject really interesting to both mathematicians and physicists. Because of the correspondence between complex numbers and two-dimensional

vectors, we might expect to be able to define the line integral of a complex function along a curve in the z-plane.

Let t be a real parameter ranging from t_A to t_B, and let $z = z(t)$ be a curve, or *contour* C, in the complex plane, with endpoints $A = z(t_A)$ and $B = z(t_B)$. (See Fig. 6.7.) Now we mark off a number of points t_i between t_A and t_B, and approximate the curve by a series of straight lines drawn from each $z(t_i)$ to $z(t_{i+1})$.

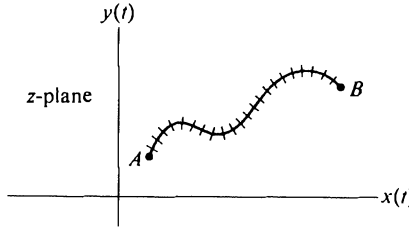

Fig. 6.7

To define the integral of a function w of a complex variable, we form the quantity

$$\lim_{|\Delta z_i| \to 0} \sum_{i=0}^{n} w(z_i)\Delta z_i \equiv \int_C w(z)\,dz\,,$$

where $\Delta z_i = z(t_{i+1}) - z(t_i)$, and $w(z_i)$ is the function evaluated at a point z_i on C between $z(t_{i+1})$ and $z(t_i)$. The sum is evaluated in the limit of an arbitrarily fine partition of the range through which the real parameter t moves as it generates the contour from A to B; that is, as $n \to \infty$, or, what is the same thing, in the limit of arbitrarily small $|\Delta z_i|$ for all i.

Writing $w(z) = u(x, y) + iv(x, y)$ and $dz = dx + i\,dy$, we have

$$\int_C w(z)\,dz = \int_C (u\,dx - v\,dy) + i\int_C (u\,dy + v\,dx)\,. \tag{6.10}$$

We can also write this in parametric form. Then

$$dx = \frac{dx}{dt}\,dt\,, \qquad dy = \frac{dy}{dt}\,dt$$

and so

$$\int_C w(z)\,dz = \int_{t_A}^{t_B}\left(u\frac{dx}{dt} - v\frac{dy}{dt}\right)dt + i\int_{t_A}^{t_B}\left(u\frac{dy}{dt} + v\frac{dx}{dt}\right)dt\,.$$

For a given contour C running from A to B we define the opposite contour, written as $-C$, to be the same curve but traversed from B to A. The integral of $w(z)$ along $-C$ is clearly given by the above equation but with t_A and t_B interchanged. Thus,

$$\int_C = -\int_{-C}\,. \tag{6.11}$$

Also it follows that

$$\int_{C_1} + \int_{C_2} = \int_{C_1+C_2} . \tag{6.12}$$

If C is a closed curve that does not intersect itself, we shall always interpret $\oint_C$ to mean the integral taken counterclockwise along the closed contour C.

Another property of the integral that we shall need very often is

$$\left| \int_C f(z)\, dz \right| \leq \int_C |f(z)|\, |dz| \leq ML , \tag{6.13}$$

where M is the maximum value of $|f(z)|$ on C, and the length of C is L. The first inequality in Eq. (6.13) is a generalization of $|z_1 + z_2| \leq |z_1| + |z_2|$, the triangle inequality; both inequalities are derived exactly as in the real case.

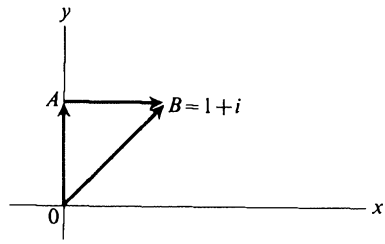

Fig. 6.8

Before presenting the general theorems on the integration of functions of a complex variable, we work out two examples of such integrals using only the results obtained so far. First, we evaluate the integral

$$I = \int_C \sin z\, dz$$

over the two paths shown in Fig. 6.8: (1) $C_1 = OB$, (2) $C_2 = OA + AB$. Since

$$\sin z = \cosh y \sin x + i \sinh y \cos x ,$$

we have, using Eq. (6.10),

$$I \equiv \int_C \sin z\, dz = \int_C [\cosh y \sin x\, dx - \sinh y \cos x\, dy]$$
$$+ i \int_C [\cosh y \sin x\, dy + \sinh y \cos x\, dx] .$$

Along the curve C_1, $x = y$. Therefore

$$I_1 = (1 + i) \int_0^1 \cosh x \sin x\, dx - (1 - i) \int_0^1 \sinh x \cos x\, dx$$
$$= (1 - \cosh 1 \cos 1) + i(\sinh 1 \sin 1) .$$

Now we compute I along C_2. Along the path from O to A, $x = 0$ and $dx = 0$,

and along the path from A to B, $y = 1$ and $dy = 0$. Therefore

$$I_2 = \int_{C_2} \sin z\, dz = -\int_0^1 \sinh y\, dy + \int_0^1 \cosh 1 \sin x\, dx + i \int_0^1 \sinh 1 \cos x\, dx$$
$$= 1 - \cosh 1 \cos 1 + i \sinh 1 \sin 1 = I_1 .$$

The integral from O to B is the same for both paths. In fact, we shall prove later that it is the same for any path whatsoever—it depends only on the two endpoints. Also, the definite integral around the closed contour consisting of C_1 and $-C_2$ (that is, C_2 traveled backward) is zero. We shall show that this result holds for any function which is analytic on and inside the closed contour.

Note that if we evaluate formally, according to the rule of real calculus,

$$I_1 = I_2 = \int_0^{1+i} \sin z\, dz = -\cos z \Big]_0^{1+i}$$
$$= 1 - \cosh 1 \cos 1 + i \sinh 1 \sin 1 ,$$

where we have used Eq. (6.9). We shall also prove this "fundamental theorem of complex calculus," which holds in any region in which the integrand is analytic.

As a second example, let us integrate the function $f(z) = z^*$ counterclockwise around the unit circle centered at the origin. The values of z on this curve are given by $z = e^{i\theta}$, $\theta = 0$ to 2π. Therefore

$$I = \oint_C z^*\, dz = \int_0^{2\pi} e^{-i\theta} i e^{i\theta}\, d\theta = 2\pi i .$$

Since $zz^* = 1$ on C, we also obtain the result

$$\oint_C \frac{1}{z}\, dz = 2\pi i .$$

Neither integral around the closed contour is zero. The reason, as we shall see, is that z^* is not analytic anywhere, and therefore not within and on C, and z^{-1} is not analytic at $z = 0$, which is within C.

Both these examples are explained by

Cauchy's Theorem. If a function $f(z)$ is analytic within and on a closed contour C, *and $f'(z)$ is continuous throughout this region*, then

$$\oint_C f(z)\, dz = 0 .$$

We shall give two proofs of this theorem.

Proof 1.

$$\oint_C f(z)\, dz = \oint_C (u\, dx - v\, dy) + i \oint_C (u\, dy + v\, dx) .$$

To evaluate the two line integrals on the right, we use Green's theorem for line integrals. It states that if the derivatives of P and Q are continuous functions

within and on a closed contour C, then

$$\oint_C (P\,dx + Q\,dy) = \int_R\int \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy,$$

where R is the surface bounded by C. By hypothesis, $f'(z)$ is continuous, so the first partial derivatives of u and v are also continuous; then Green's theorem yields

$$\oint_C (u\,dx - v\,dy) + i\oint_C (u\,dy + v\,dx)$$
$$= \int_R\int \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) dx\,dy + i\int_R\int \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx\,dy\,.$$

But since the Cauchy-Riemann equations hold, the integrands above all vanish. Therefore

$$\oint_C f(z)\,dz = 0\,. \qquad \text{QED}$$

Proof 2. Cauchy's theorem may also be proved if we use Stokes's theorem, which is closely related to Green's theorem and is perhaps more familiar. We write

$$\oint_C f(z)\,dz = \oint_C \mathbf{F}\cdot d\mathbf{l} + i\oint_C \mathbf{G}\cdot d\mathbf{l}\,,$$

where

$$\mathbf{F} = u\mathbf{i} - v\mathbf{j}\,, \qquad \mathbf{G} = v\mathbf{i} + u\mathbf{j}\,, \qquad \text{and} \qquad d\mathbf{l} = dx\mathbf{i} + dy\mathbf{j}\,.$$

Let S be the region interior to and including C. Since the Cauchy-Riemann conditions hold throughout S, it follows that $(\nabla \times \mathbf{F})_z = 0$ and $(\nabla \times \mathbf{G})_z = 0$ throughout S, where the subscript z identifies the $\mathbf{k}$ component of the curl; for example,

$$(\nabla \times \mathbf{F})_z = \frac{\partial(-v)}{\partial x} - \frac{\partial u}{\partial y} = 0\,,$$

by virtue of the Cauchy-Riemann equations. Now, using Stokes's theorem, the validity of which depends on the continuity of the four first partials of u and v,

$$\oint_C f(z)\,dz = \oint_C \mathbf{F}\cdot d\mathbf{l} + i\oint_C \mathbf{G}\cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{F})\cdot d\mathbf{S} + i\int_S (\nabla \times \mathbf{G})\cdot d\mathbf{S}$$
$$= \int_S (\nabla \times \mathbf{F})_z\,dS + i\int_S (\nabla \times \mathbf{G})_z\,dS = 0\,,$$

where $d\mathbf{S} = \mathbf{k}\,dS$. QED

It is possible to prove Cauchy's theorem without assuming the continuity of $f'(z)$. This is because any function which is analytic in a region *necessarily* has a continuous derivative. In fact, we shall prove that an analytic function has derivatives of all orders, and therefore all its derivatives are continuous, the continuity of the nth derivative being a consequence of the existence of the deriva-

tive of order $n + 1$. But we shall only be able to establish this result on higher derivatives *after* we have shown that the continuity of $f'(z)$ is not needed in the proof of Cauchy's theorem.

This relaxation or weakening of the hypotheses under which

$$\oint_C f(z)\, dz = 0$$

is therefore of the utmost importance—it is, in fact, the centerpiece of the theory of analytic functions. Some authors (never mathematicians) define an analytic function as a differentiable function *with a continuous derivative*. Then the central result of the theory follows trivially, as we have seen in the previous theorem. But this is a mathematical fraud of cosmic proportions. It was Goursat who first proved that the condition that $f'(z)$ be continuous is superfluous. It is Goursat's result that really distinguishes the theory of integration of a function of a complex variable from the theory of line integrals in the real plane. Although the theorem is often simply called Cauchy's theorem, it is the "-Goursat" half that gives it real mathematical power. In our proof we follow the presentations of Franklin and of Knopp.

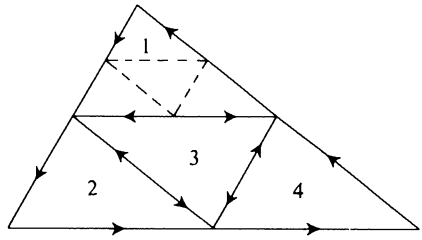

Fig. 6.9

Cauchy-Goursat Theorem. If a function $f(z)$ is analytic within and on a closed contour C, then $\oint_C f(z)\, dz = 0$.

Proof. We shall first prove the theorem for a triangular region; then it is very easily extended to an arbitrary region. Let R denote the closed region consisting of the points interior to and on the triangle bounded by the closed contour T of total length L. Since $f(z)$ is analytic in R, $f'(z)$ exists throughout R, and therefore, $f(z)$ is continuous in R. We now begin subdividing R into smaller triangles as shown in Fig. 6.9. Each subtriangle is similar to the original triangle but its sides (and perimeter) are only one-half as long. The boundaries of the subtriangles are denoted by T_i $(i = 1, 2, 3, 4)$. Clearly

$$\oint_T f(z)\, dz = \sum_{i=1}^{4} \oint_{T_i} f(z)\, dz\,,$$

where all contours are traversed in a counterclockwise direction. All three of

the boundaries of triangle 3 in Fig. 6.9 are traversed in both directions and therefore cancel out. Applying the triangle inequality to this equation, we obtain

$$\left|\oint_T f(z)\,dz\right| \leq \sum_{i=1}^{4} \left|\oint_{T_i} f(z)\,dz\right| .$$

The object of the proof is to show that the quantity on the left is arbitrarily small.

Now let C_1 denote the triangle which contributes the largest term to the above sum. Then we have

$$\left|\oint_T f(z)\,dz\right| \leq 4\left|\oint_{C_1} f(z)\,dz\right| ,$$

where the length of $C_1 \equiv L_1 = L/2$. We now repeat this process on the subtriangle bounded by C_1. That is, we find a contour C_2, bounding a "sub-subtriangle," such that

$$\left|\oint_{C_1} f(z)\,dz\right| \leq 4\left|\oint_{C_2} f(z)\,dz\right| ,$$

where the length of $C_2 \equiv L_2 = L_1/2 = L/2^2$. If the subdivision is repeated n times, we obtain a nested sequence of triangular contours C_n, such that

$$\left|\oint_T f(z)\,dz\right| \leq 4^n\left|\oint_{C_n} f(z)\,dz\right| , \tag{6.14}$$

where the length of $C_n \equiv L_n = L/2^n$. In order to finish the proof, we have to show that

$$\left|\oint_{C_n} f(z)\,dz\right|$$

is decreasing with n more rapidly than 4^n is increasing.

We let R_n denote the closed region consisting of C_n and the interior points of the (sub)$^{n\text{th}}$-triangle bounded by C_n. Clearly, each point of the region R_{n+1} is a point of R_n, and as n goes to infinity this nested sequence of closed sets closes down on a single point z_0 which is in each R_n, and R itself. (If R is the continent of Africa and we select R_n as that subregion which contains the biggest lion in Africa, we have an algorithm for capturing a big lion—we simply build a cage about the point z_0.)

Since $f'(z)$ exists, it follows by definition that for any $\epsilon > 0$, there exists a δ, such that when $0 < |z - z_0| < \delta$,

$$\left|\frac{f(z) - f(z_0)}{z - z_0} - f'(z_0)\right| \leq \epsilon .$$

Now consider the function $g(z)$ defined by

$$g(z) = \begin{cases} \dfrac{f(z) - f(z_0)}{z - z_0} - f'(z_0) & \text{for } z \neq z_0 , \\ 0 & \text{for } z = z_0 . \end{cases}$$

Note that $|g(z)| \leq \epsilon$ if $|z - z_0| < \delta$; $g(z)$ is therefore continuous at $z = z_0$.

Now $f(z)$ is given for all z in R by

$$f(z) = f(z_0) + (z - z_0)f'(z_0) + (z - z_0)g(z);$$

we use this relation to evaluate $\oint_{C_n} f(z)\,dz$. The first two terms are entire functions of z (z_0 is constant) with *derivatives* that are *continuous* everywhere. Therefore we may apply the earlier version of Cauchy's theorem to deduce that

$$\oint_{C_i} [f(z_0) + (z - z_0)f'(z_0)]\,dz = 0\,, \qquad \text{for all } i\,.$$

Therefore

$$\oint_{C_n} f(z)\,dz = \oint_{C_n} (z - z_0)g(z)\,dz\,.$$

We now subdivide enough times (i.e., we choose n large enough) so that $2^n > L/\delta$. Then $L_n = L/2^n < \delta$. Furthermore, for any point z on C_n, $|z - z_0| < L_n < \delta$, since z_0 is inside R_n, and the distance from any interior point to any point on the boundary of a triangle is clearly less than the perimeter of the triangle. Therefore, since $|z - z_0| < \delta$, $|g(z)| < \epsilon$, and $|(z - z_0)g(z)| \leq L_n\epsilon$. Consequently,

$$\left|\oint_{C_n} f(z)\,dz\right| = \left|\oint_{C_n} (z - z_0)g(z)\,dz\right| \leq L_n^2\epsilon = \epsilon(L/2^n)^2 = \epsilon L^2/4^n\,,$$

where we have used Eq. (6.13). It now follows from Eq. (6.14) that

$$\left|\oint_T f(z)\,dz\right| \leq \epsilon L^2\,.$$

Since L is the fixed finite perimeter of the triangular region R, and ϵ is arbitrary, we can make the quantity ϵL^2 smaller than any preassigned number ϵ'. Thus

$$\left|\oint_T f(z)\,dz\right| = 0\,, \qquad \text{and hence} \qquad \oint_T f(z)\,dz = 0\,.$$

This proves the Cauchy-Goursat theorem for triangular contours.

We shall not give a formal proof of the extension of this result to arbitrary regions, because the method and result are simple and clear. Given an arbitrary C, we inscribe a polygon in C. Any polygon may be decomposed into a sum of triangles, so we know the theorem holds for polygons of any number of sides. It is clear that the difference

$$\left|\oint_C f(z)\,dz - \oint_P f(z)\,dz\right|\,,$$

where P is the perimeter of the polygon inscribed in C, can be made arbitrarily small by simply choosing a polygon with a sufficiently large number of sides. This establishes the Cauchy-Goursat theorem for a region of arbitrary shape.

Throughout the proof we have tacitly assumed that the region R is a *simply-connected* region. This means that any closed contour in R encloses only points belonging to R. Suppose, however, that R were a region with one or more subregions "punched out." Then it would be possible to construct curves around

these holes in such a way that the curves would lie entirely in R, but enclose points *not* belonging to R. Such regions are called *multiply-connected*. Cauchy's theorem does not hold for arbitrary contours in multiply-connected regions.

6.4 CONSEQUENCES OF CAUCHY'S THEOREM

The hardest work is behind us; we turn now to an examination of some of the main consequences of Cauchy's theorem.

Path Independence

We first prove that if $f(z)$ is analytic in the region R and C_1 and C_2 lie in R and have the same endpoints, then

$$\int_{C_1} f\,dz = \int_{C_2} f\,dz\,.$$

The proof follows immediately by applying Cauchy's theorem to the closed contour consisting of C_2 and $-C_1$ as shown in Fig. 6.10;

$$\int_{C_2} + \int_{-C_1} = 0 \Longrightarrow \int_{C_2} = -\int_{-C_1} = \int_{C_1}$$

by Eq. (6.11).

Fundamental Theorem of Calculus

From our discussion of path independence it follows that the equation

$$F(z) \equiv \int_{z_0}^{z} f(z')\,dz'$$

defines a unique function of z if $f(z')$ is analytic throughout the region containing the path between z_0 and z.

Theorem. $F(z)$ is analytic and $F'(z) = f(z)$.

Proof.

$$F(z + \Delta z) - F(z) = \int_{z}^{z+\Delta z} f(z')\,dz'\,,$$

where the path from z to $(z + \Delta z)$ may be taken to be a straight line. We can write

$$f(z) = \frac{f(z)}{\Delta z}\int_{z}^{z+\Delta z} dz' = \frac{1}{\Delta z}\int_{z}^{z+\Delta z} f(z)\,dz'\,,$$

and it follows that

$$\frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) = \frac{1}{\Delta z}\int_{z}^{z+\Delta z} [f(z') - f(z)]\,dz'\,.$$

Now $f(z)$ is continuous because it is analytic; therefore, for all $\epsilon > 0$, there exists a $\delta > 0$ such that if $|z' - z| < \delta$, then $|f(z') - f(z)| < \epsilon$. Now take $0 < |\Delta z| < \delta$. Then

$$\left|\frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z)\right| < \epsilon\,\frac{1}{|\Delta z|}\int_{z}^{z+\Delta z} |dz'| = \epsilon\,.$$

That is,

$$F'(z) \equiv \lim_{\Delta z \to 0} \frac{F(z + \Delta z) - F(z)}{\Delta z} = f(z) ,$$

so $F(z)$ is analytic and its derivative is $f(z)$.

Thus the integral $F(z)$ of an analytic function $f(z)$ is an analytic function of its upper limit, provided the path of integration is confined to a region R within which the integrand is analytic. The fundamental theorem of calculus follows immediately from this result.

$$\int_a^b f(z)\, dz = \int_{z_0}^b f(z)\, dz - \int_{z_0}^a f(z)\, dz = F(b) - F(a) ,$$

where a and b are points in R, and $F'(z) = f(z)$, that is, $F(z)$ is an antiderivative of $f(z)$. We have already noticed that this method of evaluating integrals worked in a special case: the integral of $\sin z$ from $a = 0$ to $b = 1 + i$ (Section 6.3).

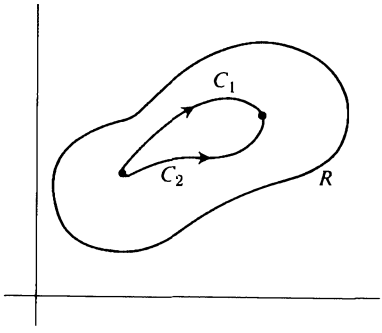

Fig. 6.10

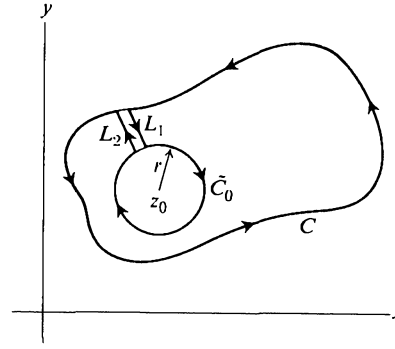

Fig. 6.11

Cauchy's Integral Formula

We now prove one of the most useful results in all mathematical physics.

Theorem. If $f(z)$ is analytic within and on a closed contour C, then for *any* point z_0, interior to C,

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0}\, dz . \tag{6.15}$$

Proof. Inside the contour C, draw a circle $\tilde{C}_0$ of radius r about z_0, and consider the contour shown in Fig. 6.11. It consists of the circle $\tilde{C}_0$ and the contour C joined by two straight line segments, L_1 and L_2, which lie arbitrarily close to each other. Let us call this entire contour C'. Now consider

$$\oint_{C'} \frac{f(z)}{z - z_0}\, dz = \oint_C \frac{f(z)}{z - z_0}\, dz + \int_{L_1} \frac{f(z)}{z - z_0}\, dz + \oint_{\tilde{C}_0} \frac{f(z)}{z - z_0}\, dz + \int_{L_2} \frac{f(z)}{z - z_0}\, dz .$$

Inside C', $f(z)/(z - z_0)$ is analytic, so by the Cauchy-Goursat theorem,

$$\oint_{C'} \frac{f(z)}{z - z_0}\, dz = 0 .$$

Now, as we bring the line segments L_1 and L_2 arbitrarily close together,

$$\int_{L_1} \frac{f(z)}{z - z_0}\, dz \to - \int_{L_2} \frac{f(z)}{z - z_0}\, dz ,$$

since the lines are traversed in opposite directions. Thus in this limit we have

$$\oint_{C'} \frac{f(z)}{z - z_0}\, dz = 0 = \oint_{C} \frac{f(z)}{z - z_0}\, dz + \oint_{\tilde{C}_0} \frac{f(z)}{z - z_0}\, dz ,$$

so that

$$\oint_{C} \frac{f(z)}{z - z_0}\, dz = - \oint_{\tilde{C}_0} \frac{f(z)}{z - z_0}\, dz .$$

At this point we note that considered as a contour in its own right, i.e., not just as a part of C', $\tilde{C}_0$ is traversed in a clockwise direction. Let us therefore define $C_0 = -\tilde{C}_0$ so that C_0 is a counterclockwise contour (as is C). Then we may write

$$\oint_{C} \frac{f(z)}{z - z_0}\, dz = \oint_{C_0} \frac{f(z)}{z - z_0}\, dz .$$

(Note that for the purposes of what we have just done, C_0 need not be a circle; it could equally well be *any* closed contour lying completely inside C and oriented in the same sense as C.) We may rewrite the last equation as

$$\oint_{C} \frac{f(z)}{z - z_0}\, dz = f(z_0) \oint_{C_0} \frac{dz}{z - z_0} + \oint_{C_0} \frac{f(z) - f(z_0)}{z - z_0}\, dz .$$

We now use the fact that C_0 is a circle to write $z - z_0 = re^{i\theta}$ on C_0. Thus the first integral on the right becomes

$$\oint_{C_0} \frac{dz}{z - z_0} = \int_0^{2\pi} \frac{ire^{i\theta} d\theta}{re^{i\theta}} = 2\pi i , \qquad \text{for all } r > 0 \text{ within } C .$$

Cauchy's formula will therefore be established if we can show that the second integral vanishes for some choice of the contour C_0. The continuity of $f(z)$ at z_0 tells us that for all $\epsilon > 0$ there exists a δ such that if $|z - z_0| \leq \delta$, then $|f(z) - f(z_0)| < \epsilon$. So by taking $r = \delta$, we satisfy the equation $|z - z_0| = \delta$, which in turn implies that

$$\left| \oint_{C_0} \frac{f(z) - f(z_0)}{z - z_0}\, dz \right| \leq \oint_{C_0} \frac{|f(z) - f(z_0)|}{|z - z_0|}\, |dz| < \frac{\epsilon}{\delta} (2\pi\delta) = 2\pi\epsilon .$$

Thus by taking r small enough, but still greater than zero, the absolute value of the integral can be made smaller than any preassigned number. Thus

$$\oint_{C} \frac{f(z)}{z - z_0}\, dz = 2\pi i\, f(z_0) .$$

This result gives us another hint of the amazingly strong inner structure of analytic functions. It means that if a function is analytic within and on a contour C, its value at every point inside C is determined by its values on the bounding curve C. There is a familiar equivalent result from electrostatics: If a real-valued function $u(x, y)$ is fixed on some boundary, and if $\nabla^2 u = 0$, then u is determined everywhere inside the boundary. An analytic function is built out of a pair of such harmonic functions. We could, if we liked, study the "theory of harmonic functions" instead of "analytic function theory".

Derivatives of Analytic Functions

Using Cauchy's integral formula, we can prove that all the derivatives of an analytic function are analytic. The corresponding result for real variables fails: a function which is once differentiable in some region is not necessarily infinitely differentiable in that region. The function $f(x) = x|x|$, for example, has as its derivative $f'(x) = 2|x|$, which is continuous everywhere; $f'(x)$ is not differentiable at the origin, however.

If we differentiate both sides of Cauchy's integral formula, interchanging the orders of integration and differentiation, we get

$$f'(z_0) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z - z_0)^2}\,dz\,. \tag{6.16}$$

Since z_0 is any point inside C, we may take it as a variable. To establish this formula in a rigorous manner, note that by using Cauchy's integral formula, we may write

$$\begin{aligned} f'(z_0) = \lim_{z_1\to z_0}\frac{f(z_1) - f(z_0)}{z_1 - z_0} &= \frac{1}{2\pi i}\lim_{z_1\to z_0}\oint_C\left[\frac{f(z)}{z - z_1} - \frac{f(z)}{z - z_0}\right]\frac{dz}{z_1 - z_0} \\ &= \frac{1}{2\pi i}\lim_{z_1\to z_0}\oint_C \frac{f(z)}{(z - z_1)(z - z_0)}\,dz\,. \end{aligned}$$

Hence

$$\begin{aligned} f'(z_0) - \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z - z_0)^2}\,dz &= \frac{1}{2\pi i}\lim_{z_1\to z_0}\oint_C f(z)\left[\frac{1}{(z - z_1)(z - z_0)} - \frac{1}{(z - z_0)^2}\right]dz \\ &= \frac{1}{2\pi i}\lim_{z_1\to z_0}(z_1 - z_0)\oint_C \frac{f(z)}{(z - z_1)(z - z_0)^2}\,dz\,. \end{aligned}$$

Calling $z_1 - z_0 = \epsilon e^{i\theta}$, we have

$$\left|f'(z_0) - \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z - z_0)^2}\,dz\right| \le \frac{1}{2\pi}\lim_{\epsilon\to 0}\epsilon\oint_C \frac{|f(z)|\,|dz|}{|(z - z_0) - \epsilon e^{i\theta}|\,|z - z_0|^2}\,.$$

Replacing $|z - z_0|$ by its minimum value, say μ, and $|f(z)|$ by its maximum value M, we obtain

$$\left|f'(z_0) - \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z - z_0)^2}\,dz\right| \le \frac{1}{2\pi}\frac{ML}{\mu^2}\lim_{\epsilon\to 0}\frac{\epsilon}{\mu - \epsilon} = 0\,,$$

where L is the length of the contour. Thus we have proved Eq. (6.16). Repeat-

ing the process, we obtain

$$f''(z_0) = \frac{2!}{2\pi i}\oint_C \frac{f(z)}{(z - z_0)^3}\,dz\,,$$

and, in general, for the nth derivative,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\oint_C \frac{f(z)}{(z - z_0)^{n+1}}\,dz\,. \tag{6.17}$$

This result is readily established by induction in the same manner we used to prove Eq. (6.16). Thus $f(z)$ has derivatives of all orders within C. The kth derivative of $f(z)$ is continuous within C because the $(k + 1)$ derivative exists. Thus if we write $f(z) = u(x, y) + iv(x, y)$, the partial derivatives of u and v of all orders are continuous whenever f is analytic. We can therefore drop in our derivation of Eq. (6.4) the *requirement* that the second partial derivatives be continuous: they are guaranteed to be continuous because f is analytic.

Liouville's Theorem

Theorem. If $f(z)$ is entire and $|f(z)|$ is bounded for all values of z, then $f(z)$ is a constant.

Proof. From Cauchy's integral formula, we have found that

$$f'(z_0) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z - z_0)^2}\,dz\,.$$

If we take C to be the circle $|z - z_0| = r_0$, then

$$\begin{aligned} |f'(z_0)| &\le \left|\frac{1}{2\pi i}\right| \oint_{C_0} \frac{|f(z)|}{|(z - z_0)^2|}\,|dz| \\ &< \frac{1}{2\pi r_0^2}\,M 2\pi r_0 = \frac{M}{r_0}\,, \end{aligned}$$

where $|f(z)| < M$ within and on C_0. Therefore $|f'(z_0)| < M/r_0$, and we may take r_0 as large as we like because $f(z)$ is entire. So taking r_0 large enough, we can make $|f'(z_0)| < \epsilon$, for any preassigned ϵ. That is, $|f'(z_0)| = 0$, which implies that $f'(z_0) = 0$ for all z_0, so $f(z_0) =$ constant. QED

Example. The entire functions $\sin z$ and $\cos z$ must not be bounded. It is clear from Eqs. (6.8) and (6.9) that they are not.

Fundamental Theorem of Algebra

This theorem, which is difficult to prove algebraically, follows easily from Liouville's theorem, and provides a remarkable tie-up between analysis and algebra. We include the proof because of its great simplicity and beauty.

Theorem. If $P(z) = a_0 + a_1 z + \cdots + a_m z^m$ is a polynomial in z of degree one or greater, then the equation $P(z) = 0$ has at least one root.

Proof. Assume the contrary, namely that $P(z) \neq 0$ for any z. Then the function $1/P(z)$ is entire. Furthermore $|1/P(z)| \to 0$ as $|z| \to \infty$ so $|1/P(z)|$ is bounded

for all z. Therefore, by Liouville's theorem, $1/P(z) = \text{const}$, a contradiction, since $P(z)$ is of degree one or greater. Hence $P(z) = 0$, for at least one value of z. QED

6.5 HILBERT TRANSFORMS AND THE CAUCHY PRINCIPAL VALUE

It is often the case that in the study of some physical system one has to deal with complex-valued functions—indices of refraction, susceptibilities, scattering amplitudes, impedances, etc.—which have a physical meaning only when the argument of the function (which might, for example, be a frequency or an energy) takes on *real* values. In many cases it is possible to obtain, from the laws governing the system, information about the general properties of such functions when the argument is *complex*; for example, it may be that the function is analytic in some region of the complex plane. Since experimental data can only be obtained for real values of the argument, it is of interest to see whether we can use general properties such as analyticity to deduce relations between real quantities of direct physical significance. The key to such a program can be found in the study of Hilbert transform pairs, which we shall investigate in this section.

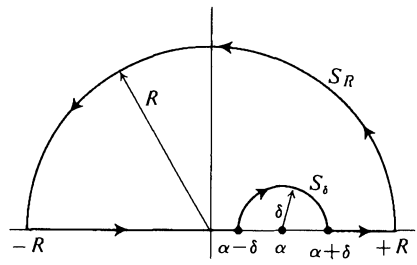

Fig. 6.12 The contour, C, used to obtain Eq. (6.18). The radius, R, of the semi-circle, S_r, may be made as large as necessary, and the radius, S, of the semi-circle, S_δ, may be made as small as we please.

Let us begin by considering a function $f(z)$, which is analytic in the upper half of the complex plane, and which is such that $|f(z)| \to 0$ as $|z| \to \infty$ in the upper half-plane. (Note that the only function which can satisfy these conditions in the *entire* plane is $f \equiv 0$). Now consider the contour integral

$$\oint_C \frac{f(z)}{z - \alpha}\, dz\,,$$

where C is the contour shown in Fig. 6.12 and α is real. By assumption, $f(z)$ is analytic within and on C; so is $1/(z - \alpha)$. Thus

$$\oint_C \frac{f(z)}{z - \alpha}\, dz = 0\,.$$

Let us break this up as follows:

$$\oint_C \frac{f(z)}{z-\alpha}\,dz \equiv \int_{-R}^{\alpha-\delta} \frac{f(x)}{x-\alpha}\,dx + \int_{S_\delta} \frac{f(z)}{z-\alpha}\,dz$$
$$+ \int_{\alpha+\delta}^{R} \frac{f(x)}{x-\alpha}\,dx + \int_{S_R} \frac{f(z)}{z-\alpha}\,dz = 0\,.$$

Here δ is the radius of the small semicircle S_δ, centered at $x = \alpha$, and R is the radius of the large semicircle S_R, centered at the origin, as shown in Fig. 6.12. The radius δ can be chosen as small as we please, and R can be chosen as large as we please. In the limit of arbitrarily small δ, the quantity

$$\int_{-R}^{\alpha-\delta} \frac{f(x)}{x-\alpha}\,dx + \int_{\alpha+\delta}^{R} \frac{f(x)}{x-\alpha}\,dx$$

is called the *principal-value integral* of $f(x)/(x-\alpha)$ and is denoted by

$$P\int_{-R}^{R} \frac{f(x)}{x-\alpha}\,dx\,.$$

We will say more about this integral below. Now, along the large semicircle S_R, we set $z = Re^{i\theta}$, so that

$$\int_{S_R} \frac{f(z)}{z-\alpha}\,dz = i\int_0^\pi \frac{f(Re^{i\theta})}{Re^{i\theta}-\alpha}\,Re^{i\theta}\,d\theta\,,$$

and hence

$$\left|\int_{S_R} \frac{f(z)}{z-\alpha}\,dz\right| \le \frac{R}{|R-\alpha|}\int_0^\pi |f(Re^{i\theta})|\,d\theta\,,$$

since $|Re^{i\theta}-\alpha| = [R^2+\alpha^2-2R\alpha\cos\theta]^{1/2} \ge [R^2+\alpha^2-2R\alpha]^{1/2} = |R-\alpha|$. But as $R\to\infty$, $|f(z)|\to 0$ and $R/|R-\alpha|\to 1$. Therefore the integral over the semicircle of radius R can be made arbitrarily small by choosing R sufficiently large. Thus we may write

$$\lim_{R\to\infty} P\int_{-R}^{R} \frac{f(x)}{x-\alpha}\,dx = -\int_{S_\delta} \frac{f(z)}{z-\alpha}\,dz = -f(\alpha)\int_{S_\delta} \frac{dz}{z-\alpha} - \int_{S_\delta} \frac{f(z)-f(\alpha)}{z-\alpha}\,dz\,,$$

where we have added and subtracted the term $\int_{S_\delta}[f(\alpha)/(z-\alpha)]\,dz$. Setting $z-\alpha = \delta e^{i\theta}$ in the first integral on the right-hand side of this equation, we find that

$$-f(\alpha)\int_{S_\delta} \frac{dz}{z-\alpha} = -if(\alpha)\int_\pi^0 d\theta = i\pi f(\alpha)\,.$$

Thus

$$\lim_{R\to\infty} P\int_{-R}^{R} \frac{f(x)}{x-\alpha}\,dx = i\pi f(\alpha) - \int_{S_\delta} \frac{f(z)-f(\alpha)}{z-\alpha}\,dz\,.$$

Since $f(z)$ is continuous at $z = \alpha$, the argument used in deriving Cauchy's

integral formula tells us that this last integral over S_δ vanishes. Hence

$$\lim_{R\to\infty} P\int_{-R}^{R} \frac{f(x)}{x-\alpha}\,dx = i\pi f(\alpha)\ .$$

For the sake of brevity, we write this simply as

$$P\int_{-\infty}^{\infty} \frac{f(x)}{x-\alpha}\,dx = i\pi f(\alpha)\ , \tag{6.18}$$

where $f(x)$ is a complex-valued function of a real variable. We may write it as

$$f(x) \equiv f_R(x) + if_I(x)\ .$$

Equating real and imaginary parts in Eq. (6.18), we get

$$f_R(\alpha) = \frac{1}{\pi} P\int_{-\infty}^{\infty} \frac{f_I(x)}{x-\alpha}\,dx\ , \tag{6.19a}$$

$$f_I(\alpha) = -\frac{1}{\pi} P\int_{-\infty}^{\infty} \frac{f_R(x)}{x-\alpha}\,dx\ . \tag{6.19b}$$

Any pair of functions which satisfy Eqs. (6.19a) and (6.19b) is called a *Hilbert transform pair*. Note that these equations tell us that if $f_I(x) \equiv 0$, then $f_R(x) \equiv 0$.

The principal-value integral is seen to be a way of avoiding singularities on a path of integration: one integrates to within δ of the singularity in question, skips over the singularity, and begins integrating again a distance δ beyond the singularity. This prescription enables one to make sense out of integrals like

$$\int_{-R}^{R} \frac{dx}{x}\ .$$

One would like this integral to be zero, since we are integrating an odd function over a symmetric domain. However, unless we insert a P in front of this integral, the singularity at the origin makes the integral meaningless. Following the prescription for principal-value integrals, we can easily evaluate the above integral. We have

$$P\int_{-R}^{R} \frac{dx}{x} = \lim_{\delta\to 0}\left[\int_{-R}^{-\delta} \frac{dx}{x} + \int_{\delta}^{R} \frac{dx}{x}\right].$$

In the first integral on the right-hand side, set $x = -y$. Then

$$P\int_{-R}^{R} \frac{dx}{x} = \lim_{\delta\to 0}\left[\int_{R}^{\delta} \frac{dy}{y} + \int_{\delta}^{R} \frac{dx}{x}\right].$$

The sum of the two integrals inside the brackets is zero, since

$$\int_{R}^{\delta} = -\int_{\delta}^{R}\ .$$

Thus

$$P\int_{-R}^{R} \frac{dx}{x} = 0\ .$$

In a similar manner, we may evaluate

$$P\int_{-R}^{R}\frac{dx}{x-a}$$

when $-R < a < R$. As above, we write

$$P\int_{-R}^{R}\frac{dx}{x-a} = \lim_{\delta\to 0}\left[\int_{-R}^{a-\delta}\frac{dx}{x-a} + \int_{a+\delta}^{R}\frac{dx}{x-a}\right].$$

Again setting $x = -y$ in the first integral on the right-hand side, we find that

$$\begin{aligned} P\int_{-R}^{R}\frac{dx}{x-a} &= \lim_{\delta\to 0}\left[\int_{R}^{\delta-a}\frac{dy}{y+a} + \ln(R-a) - \ln\delta\right] \\ &= \lim_{\delta\to 0}[\ln\delta - \ln(R+a) + \ln(R-a) - \ln\delta]\,. \end{aligned}$$

Thus

$$P\int_{-R}^{R}\frac{dx}{x-a} = \ln\left(\frac{R-a}{R+a}\right), \qquad -R < a < R\,. \tag{6.20}$$

For the case

$$P\int_{-R}^{R}\frac{f(x)}{x-a},$$

the result of Eq. (6.20) leads to

$$P\int_{-R}^{R}\frac{f(x)}{x-a}\,dx = P\int_{-R}^{R}\frac{f(a)}{x-a}\,dx + P\int_{-R}^{R}\frac{f(x)-f(a)}{x-a}\,dx\,,$$

or

$$P\int_{-R}^{R}\frac{f(x)}{x-a}\,dx = f(a)\ln\left(\frac{R-a}{R+a}\right) + P\int_{-R}^{R}\frac{f(x)-f(a)}{x-a}\,dx\,. \tag{6.21}$$

It will often happen that the second integral on the right-hand side of Eq. (6.21) will not be singular at $x = a$ [for example, this will be the case if $f(x)$ is differentiable at $x = a$] so the P symbol there can be dropped. We leave it to the reader to obtain the closely related result:

$$P\int_{0}^{R}\frac{f(x)}{x^2-a^2}\,dx = f(a)\,\frac{1}{2a}\ln\left(\frac{R-a}{R+a}\right) + P\int_{0}^{R}\frac{f(x)-f(a)}{x^2-a^2}\,dx\,. \tag{6.22}$$

To illustrate the use of the principal-value method and also the use of Eqs. (6.19a) and (6.19b), we consider the function $f(z) = 1/(z+i)$. This function satisfies all the hypotheses made in deriving Eqs. (6.19a) and (6.19b). We see that

$$f_R(x) = \frac{x}{x^2+1}\,, \qquad f_I(x) = -\frac{1}{x^2+1}\,.$$

Let us examine Eq. (6.19b). We write it as

$$\lim_{R\to\infty} P\int_{-R}^{R}\frac{f_R(x)}{x-\alpha}\,dx = -\pi f_I(\alpha)\,.$$

We want to see if $f(z) = 1/(z + i)$ satisfies this relation. Using Eq. (6.21), we have

$$\lim_{R\to\infty} P\int_{-R}^{R} \frac{f_R(x)}{x-\alpha}\,dx = \lim_{R\to\infty}\left[f_R(\alpha)\ln\left(\frac{R-\alpha}{R+\alpha}\right) + P\int_{-R}^{R}\frac{f_R(x)-f_R(\alpha)}{x-\alpha}\,dx\right]. \tag{6.23}$$

Now, since $f_R(x) = x/(x^2+1)$, we find that

$$\frac{f_R(x)-f_R(\alpha)}{x-\alpha} = \frac{1-\alpha x}{(\alpha^2+1)(x^2+1)},$$

so we may drop the principal-value sign on the right-hand side of Eq. (6.23). Also,

$$\lim_{R\to\infty}\ln\left(\frac{R-\alpha}{R+\alpha}\right) = 0,$$

so Eq. (6.23) becomes

$$\lim_{R\to\infty} P\int_{-R}^{R}\frac{f_R(x)}{x-\alpha}\,dx = \frac{1}{\alpha^2+1}\lim_{R\to\infty}\int_{-R}^{R}\frac{1-\alpha x}{x^2+1}\,dx = \frac{1}{\alpha^2+1}\lim_{R\to\infty}\int_{-R}^{R}\frac{dx}{x^2+1},$$

since $x/(x^2+1)$ is an odd function. Thus

$$\lim_{R\to\infty} P\int_{-R}^{R}\frac{f_R(x)}{x-\alpha}\,dx = \frac{2}{\alpha^2+1}\lim_{R\to\infty}\tan^{-1}R = \frac{\pi}{\alpha^2+1} = -\pi f_I(\alpha),$$

so Eq. (6.19b) is indeed satisfied. We leave it to the reader to show that Eq. (6.19a) is also satisfied. Clearly Eq. (6.21) is very useful in conjunction with Eqs. (6.19a) and (6.19b) because under the assumptions made in deriving these equations, $f(z)$ is differentiable at $z = \alpha$. Thus

$$f_R(\alpha) = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{f_I(x)-f_I(\alpha)}{x-\alpha}\,dx, \tag{6.24a}$$

$$f_I(\alpha) = -\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{f_R(x)-f_R(\alpha)}{x-\alpha}\,dx, \tag{6.24b}$$

where, in case of any ambiguity,

$$\int_{-\infty}^{\infty} \equiv \lim_{R\to\infty}\int_{-R}^{R}.$$

These methods can sometimes be used to evaluate real definite integrals. For example, consider the function $f(z) = e^{iz}$. This function is analytic everywhere, and if we write $z = Re^{i\theta}$, then $|f(z)| \to 0$ as $R \to \infty$ for all θ such that $0 < \theta < \pi$. This is not quite what we used above to show that the integral around a large semicircle of $f(z)/(z-\alpha)$ vanishes (although it would be the same if we had $0 \le \theta \le \pi$). However, the reader can show for $f(z) = e^{iz}$ that the contribution from the large semicircle vanishes and Eqs. (6.19a) and (6.19b) are satisfied. In this case, $f_R(x) = \cos x$ and $f_I(x) = \sin x$, so using Eq.

(6.24a), we obtain

$$\cos\alpha = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin x - \sin\alpha}{x - \alpha}\,dx\,.$$

Since $\sin x - \sin\alpha = 2\sin\frac{1}{2}(x-\alpha)\cos\frac{1}{2}(x+\alpha)$, we see that indeed there is no singularity of the integrand at $x = \alpha$. For the special case $\alpha = 0$, we find that

$$1 = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin x}{x}\,dx\,,$$

that is,

$$\int_{-\infty}^{\infty} \frac{\sin x}{x}\,dx = \pi\,.$$

From this result, we also obtain by symmetry

$$\int_{0}^{\infty} \frac{\sin x}{x}\,dx = \frac{\pi}{2}\,.$$

Here we see that the oscillations of $\sin x$, when x is large, make the integral converge, even though

$$\int_{1}^{\infty} \frac{1}{x}\,dx$$

diverges. This is analogous to the fact that the *alternating* series

$$\sum_{n=1}^{\infty} (-1)^{n+1}\frac{1}{n}\ (= \ln 2)$$

converges, whereas $\sum_{n=1}^{\infty}(1/n)$ diverges.

6.6 AN INTRODUCTION TO DISPERSION RELATIONS

As mathematical results, the equations derived in the previous section are interesting in their own right. However, a scientist naturally wants to know if there are any physical systems to which these results can be applied. What we shall now show is that under fairly broad physically motivated assumptions, one can find physical quantities which possess the analytic properties necessary for them to satisfy a Hilbert transform relation. In our detailed applications, we will focus our attention on electromagnetic theory, but many of our results will be more general than this.

We begin by considering a physical system for which an input, $I(t)$, is related to a response, $R(t)$, in the following *linear* manner:

$$R(t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} G(t-t')I(t')\,dt'\,. \tag{6.25}$$

For example, $I(t')$ might be the electric field at a time t', and $R(t)$ might be the resulting polarization field at time t. We have assumed that G depends only on

$t - t'$ because we want the system to respond to a sharp input at t_0, $I(t') = I_0\delta(t' - t_0)$, in the same way it would respond to a sharp input at $t_0 + \tau$, that is, at a time τ later. For the first case, we have

$$R_1(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} G(t - t') I_0 \delta(t' - t_0)\, dt' = \frac{1}{\sqrt{2\pi}} I_0 G(t - t_0) .$$

For the second case, we have

$$R_2(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} G(t - t') I_0 \delta(t' - t_0 - \tau)\, dt' = \frac{1}{\sqrt{2\pi}} I_0 G(t - t_0 - \tau) ,$$

or, in other words,

$$R_2(t + \tau) = \frac{1}{\sqrt{2\pi}} I_0 G(t - t_0) = R_1(t) .$$

Thus if we shift the input by τ, then we shift the response by τ.

Now, what can we say about $G(\tau)$ on general physical grounds? First we see that an input at t should not give rise to a response at times prior to t, that is, $G(\tau) = 0$ for $\tau < 0$, so

$$R(t) = \int_{-\infty}^{t} G(t - t') I(t')\, dt' ,$$

which shows that the response at t is the weighted linear superposition of all inputs *prior* to t. This is the *causality requirement*. The possibility that $G(\tau)$ is singular for any finite τ is excluded since the response from a sharp input, $I(t') = I_0\delta(t' - t_0)$, is

$$R(t) = \frac{1}{\sqrt{2\pi}} I_0 G(t - t_0) , \qquad t > t_0 ,$$

and since on physical grounds we require that this response always be finite, $G(\tau)$ is finite for all τ. Furthermore, we make the physically reasonable assumption that the effect of an input in the *remote* past does not appreciably influence the present. This may be stated as the requirement that $G(\tau) \to 0$ as $\tau \to \infty$ since, from the previous equation, it amounts to the assumption that the response to any impulse dies down after a sufficiently long time (i.e., any system has some dissipative mechanism).

Now consider the Fourier transform of Eq. (6.25). Using the convolution theorem (see Section 5.7), we find that

$$r(\omega) = g(\omega) i(\omega) ,$$

where

$$r(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} R(t) e^{i\omega t}\, dt , \qquad g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} G(t) e^{i\omega t}\, dt ,$$

$$i(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} I(t) e^{i\omega t}\, dt .$$

In electromagnetic theory, where I is the applied electric field $\mathbf{E}$, and R is the polarization field $\mathbf{P}$, it is customary to denote $g(\omega)$ by $\chi(\omega)$, which is referred

to as the electric susceptibility. Thus

$$P(\omega) = \chi(\omega)E(\omega) .$$

By assuming that $G(\tau)$ satisfies

$$\int_0^\infty |G(\tau)|\, d\tau < \infty ,$$

we can guarantee the existence of a bounded $g(\omega)$ for all ω. We may now summarize our physically motivated assumptions on $G(\tau)$:

a) $G(\tau)$ is bounded for all τ;
b) $|G(\tau)|$ is integrable, so $G(\tau) \to 0$ faster than $1/\tau$ as $\tau \to \infty$;
c) $G(\tau) = 0$ for $\tau < 0$.

We may remark that (a) and (b) taken together imply that $G(\tau)$ is square integrable and hence (see Section 9.6) $g(\omega)$ is square integrable.

We now want to show that we can extend $g(\omega)$ into the complex z-plane in such a way that $g(z)$ satisfies the conditions under which we derived the Hilbert transform pair [Eqs. (6.19a) and (6.19b)] of the previous section. First, since $G(\tau) = 0$ for $\tau < 0$, we write

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_0^\infty G(t)e^{i\omega t}\, dt .$$

We extend this relation into the complex plane by using the definition

$$\begin{aligned} g(z) &\equiv \frac{1}{\sqrt{2\pi}} \int_0^\infty G(t)e^{izt}\, dt \\ &= \frac{1}{\sqrt{2\pi}} \int_0^\infty G(t)e^{i\omega t}e^{-\omega' t}\, dt , \end{aligned}$$

where we have written $z = \omega + i\omega'$. We now restrict our attention to the upper half-plane ($\omega' > 0$) where, because of the causality requirement given in assumption (c) above, ($t > 0$ in the above integral), the term $e^{-\omega' t}$ is a *decaying* exponential. For $0 < \theta < \pi$, we have

$$|g(z)| \leq \frac{1}{\sqrt{2\pi}} M_G \int_0^\infty e^{-[|z| \sin \theta]t}\, dt ,$$

where we have replaced $G(t)$ by its maximum value, M_G [assumption (a) above]. Thus

$$|g(z)| \leq \frac{M_G}{\sqrt{2\pi}\, |z| \sin \theta} ,$$

which tends to zero when $|z| \to \infty$. For $\theta = 0$ or π, we have

$$g(\omega, \omega' = 0) = \frac{1}{\sqrt{2\pi}} \int_0^\infty G(t)e^{i\omega t}\, dt .$$

Since $G(t)$ is square integrable, so is $g(\omega, \omega' = 0)$ as a function of ω (see Section 9.6), and hence $|g(\omega, \omega' = 0)|$ tends to zero as $\omega \to \infty$. Thus in any direction in the upper half-plane, $|g(z)| \to 0$ as $|z| \to \infty$.